
Chapter 3

Single-Stage Amplifiers

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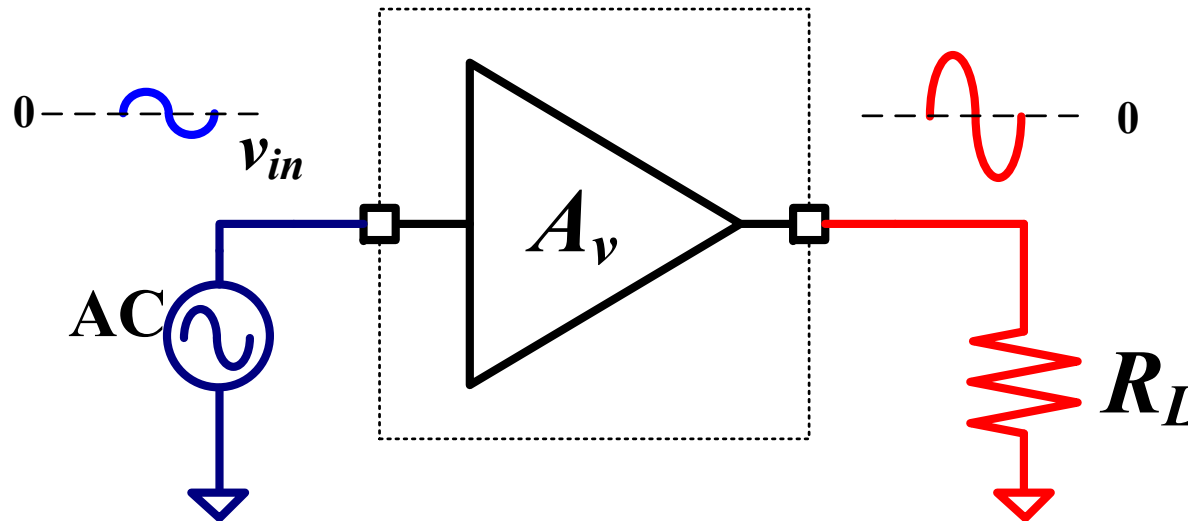
Outline

- **Fundamental Concepts**
- Common-Source Stage
- Source Follower
- Common-Gate Stage
- Cascode Stage



Why do we need amplifiers?

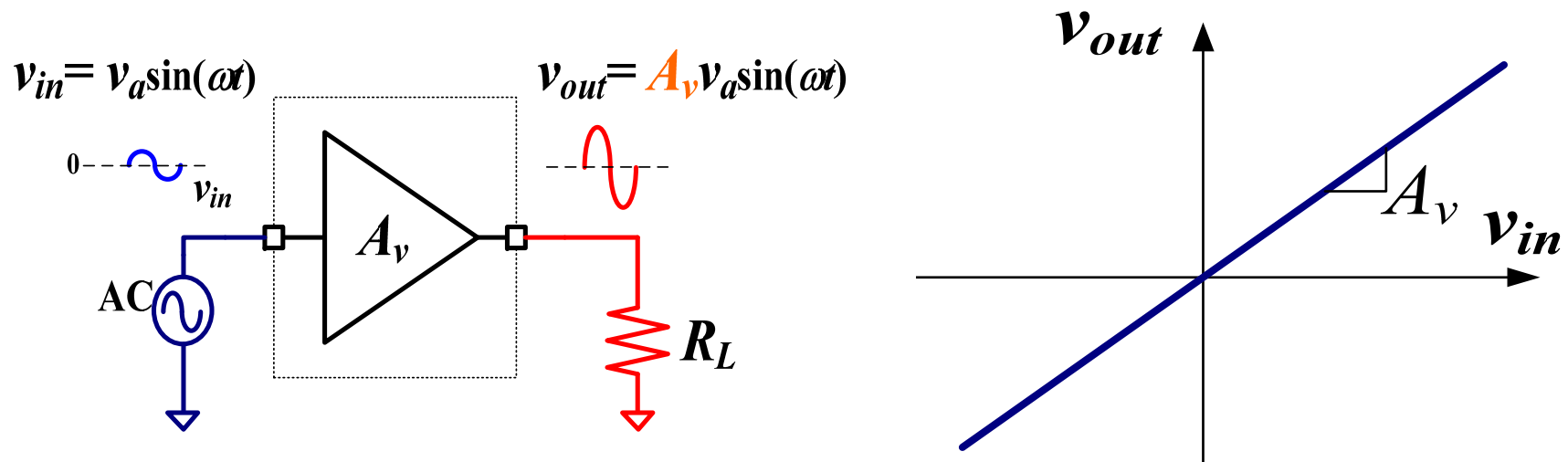
- Amplification is an essential function for analog circuits
 - Most natural analog signals are too small to process
 - We need enough **signal to noise ratio (SNR)**
 - Amplification plays critical role in feedback systems



Ideal voltage amplifier

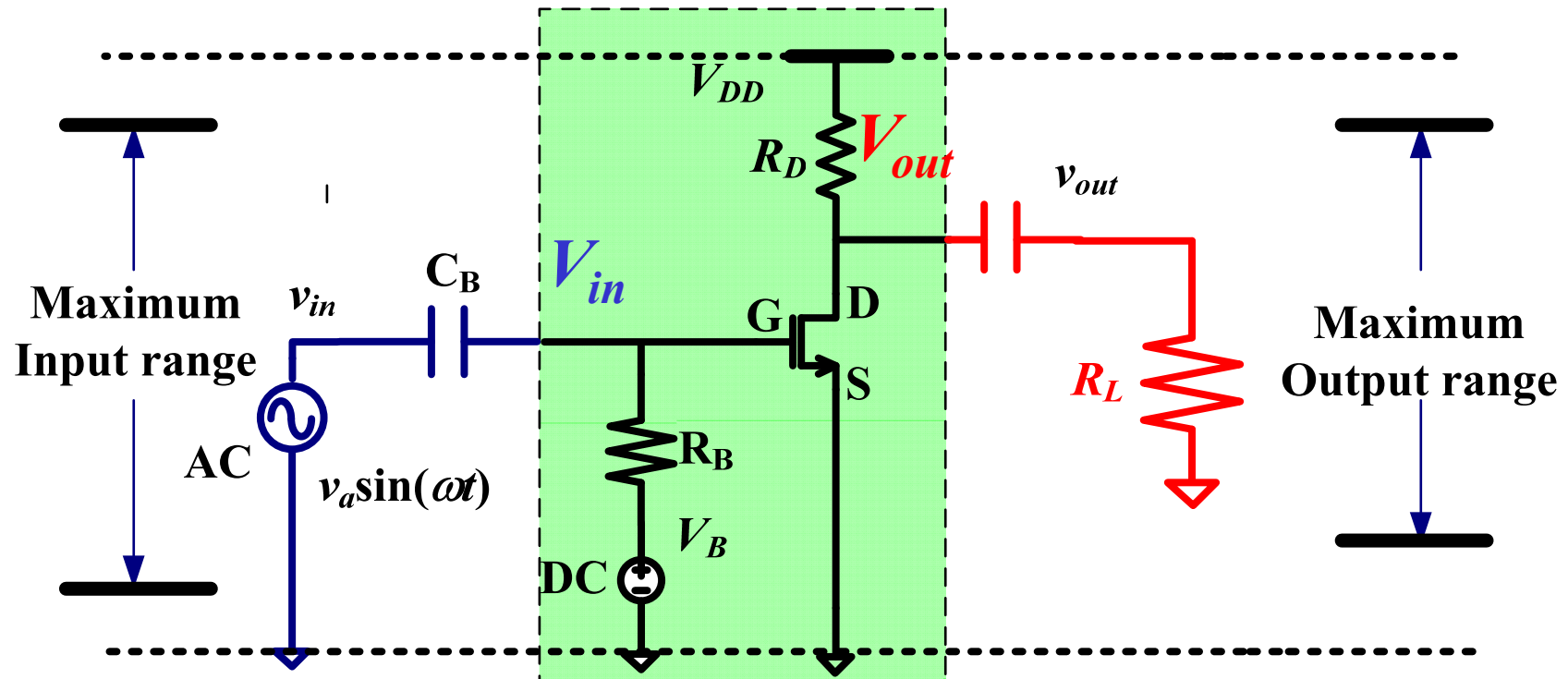
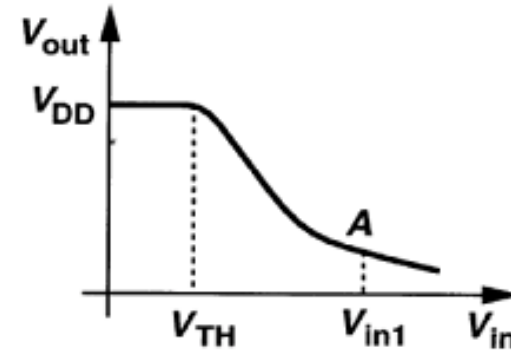
- Linear (no amplitude nor frequency distortions)
 - Infinite signal amplitude range
 - Infinite frequency range (f can be DC to ∞)
- Infinite input resistance (without loading for the previous stage)
- Infinite driving capability (R_L can be arbitrarily small)

$$v_{out} = A_v \cdot v_{in}$$



Real CMOS amplifier: An example

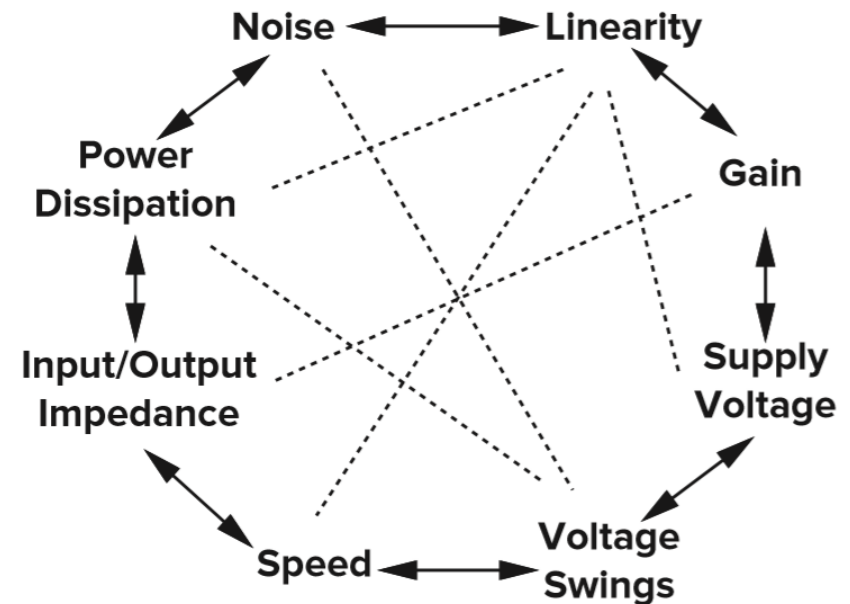
- Finite signal range
- Finite frequency range
- Distortion (nonlinear)



What do we care about an amplifier?

- Supply Voltage (V_{DD})
- DC range (Swing)
- Small signal voltage gain (A_v)
- Input/output resistance, r_{in} , r_{out} (or impedance)
- Frequency response (more later)
- Noise
- Linearity
- Power dissipation (biasing current)
- Layout area (sizes of transistors)

Analog design Octagon



- Most parameters trade with each other
 - Analog design: multi-dimensional optimization problem



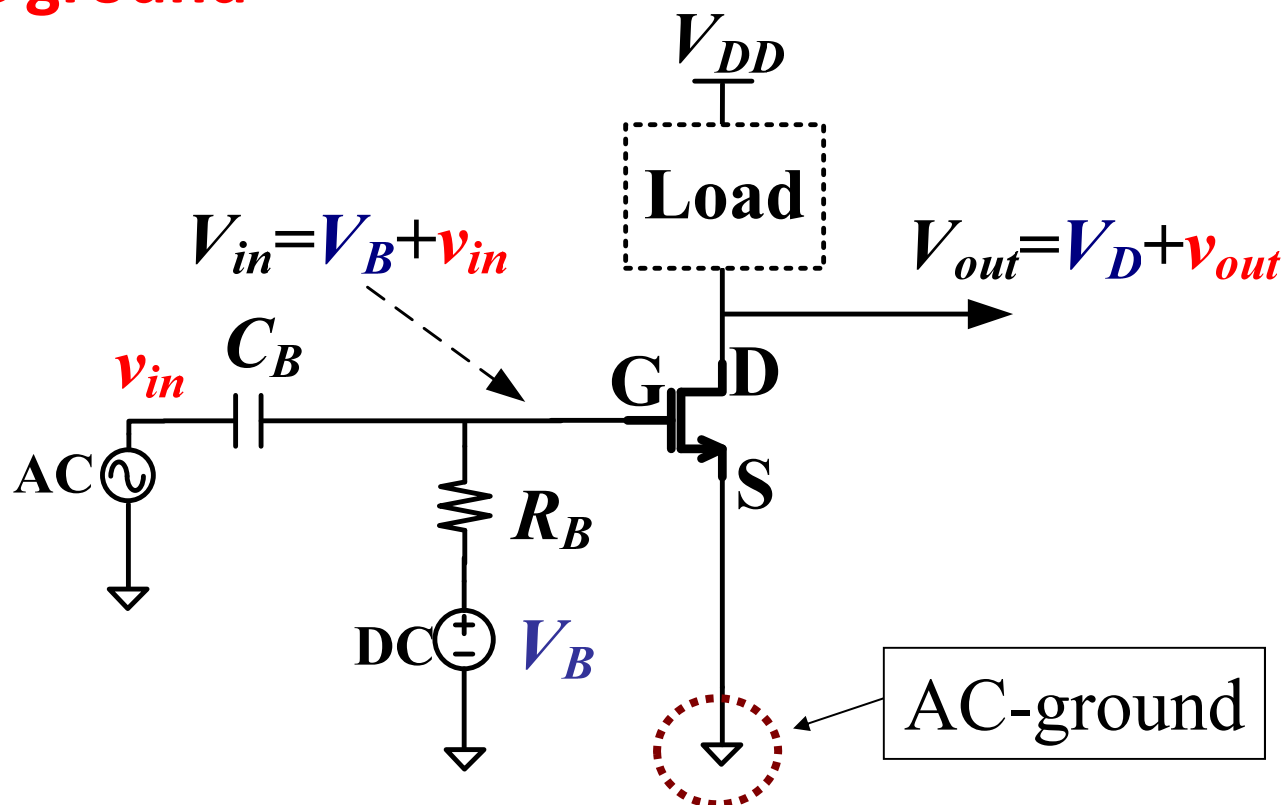
Outline

- Basic concepts
- **Common-source stage**
 - **Resistive load**
 - Diode-connected load
 - Current source load
 - Versatile analysis method for CS stage
 - CS stage with source degeneration
- Source Follower
- Common-Gate stage
- Cascode stage



What is a **common-source** stage?

- A single-transistor amplifier with **source connected to AC ground**



- What device can we put into the load box?

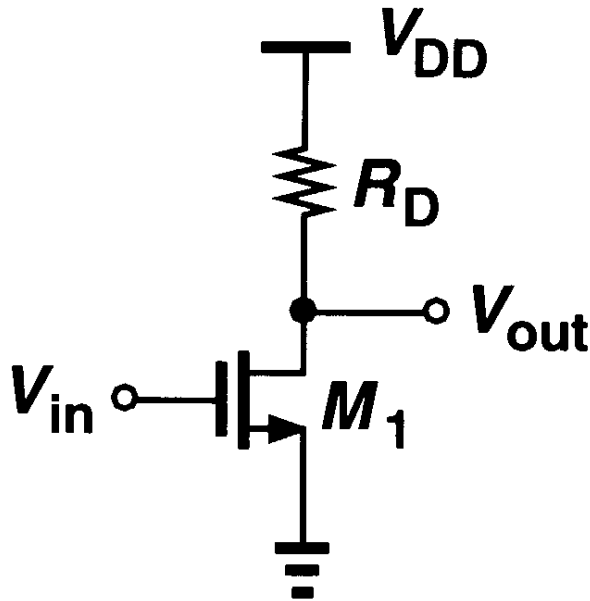
CS-stage with resistive load

- Large-signal analysis

- $V_{in} < V_{TH}$, cut off

$$V_{out} = V_{DD}$$

- $V_{in} - V_{TH} \leq V_{out}$, saturation

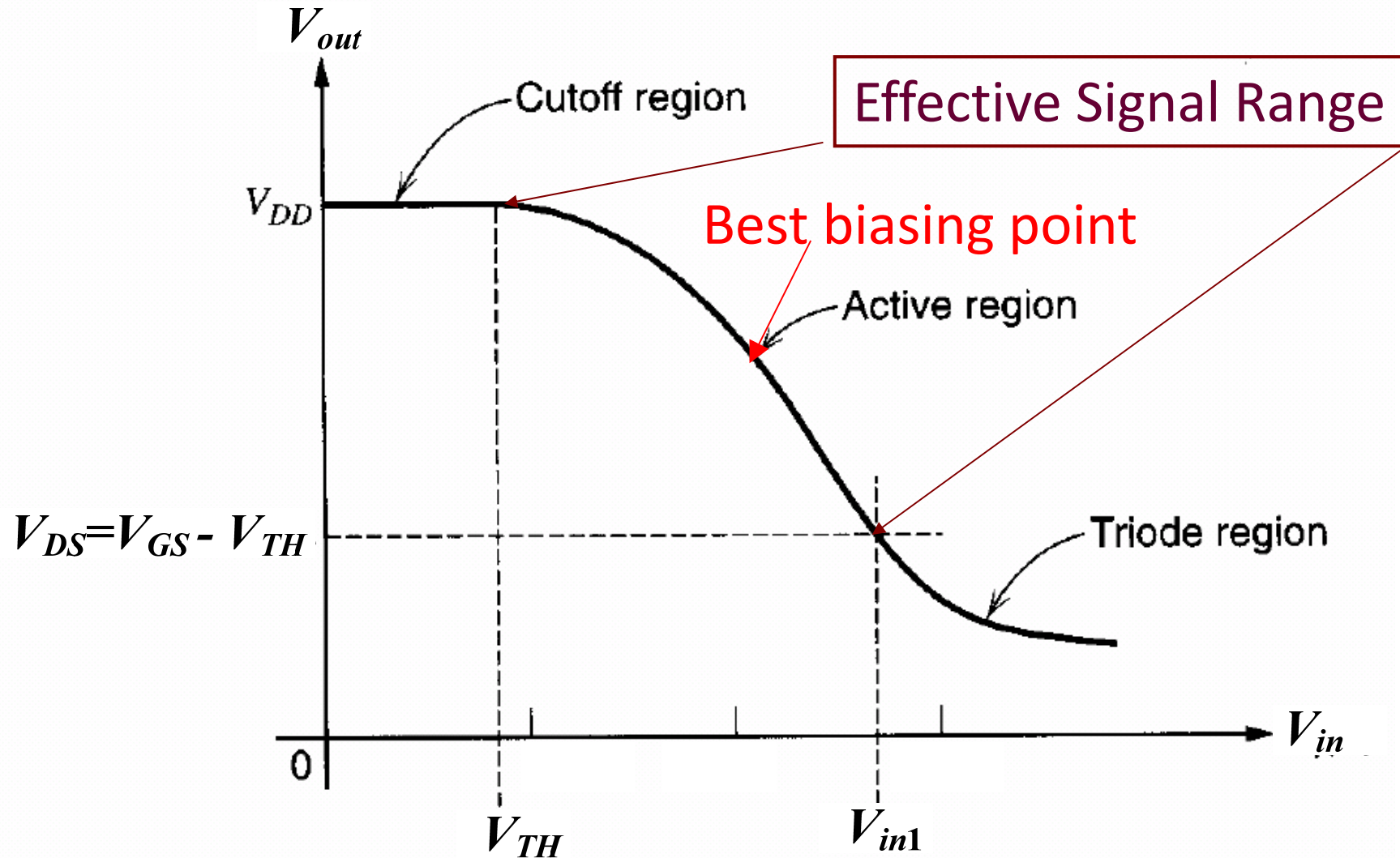


$$\begin{aligned} V_{out} &= V_{DD} - I_d \cdot R_D \\ &= V_{DD} - \frac{\mu_n C_{ox}}{2} \frac{W}{L} (V_{in} - V_{TH})^2 \cdot R_D \end{aligned}$$

- $V_{in} - V_{TH} > V_{out}$, triode



Large- signal input/output curve



Small-signal gain in the active region

- Method 1: calculate the slope of the large-signal input/output curve

$$V_{out} = V_{DD} - \frac{\mu_n C_{ox}}{2} \frac{W}{L} (V_{in} - V_{TH})^2 \cdot R_D$$

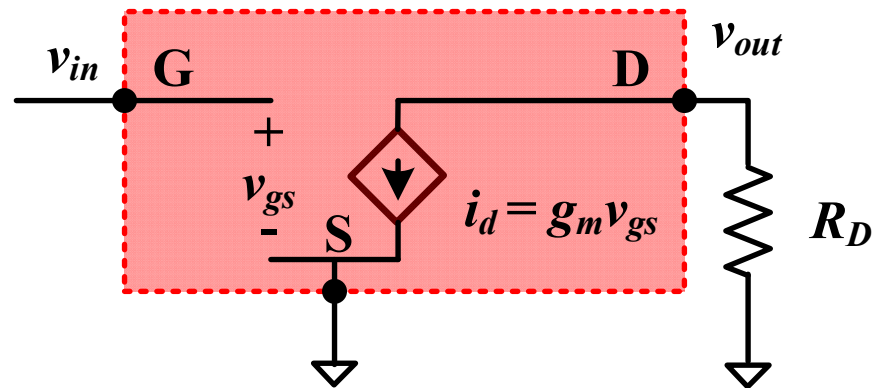
$$\longrightarrow A_v = \frac{\partial V_{out}}{\partial V_{in}} = - \overbrace{\mu_n C_{ox} \frac{W}{L} (V_{in} - V_{TH})}^{g_m} \cdot R_D$$

$$A_v = \frac{\partial V_{out}}{\partial V_{in}} = -g_m \cdot R_D$$

- Note: Different V_{in} results in different A_v , why?



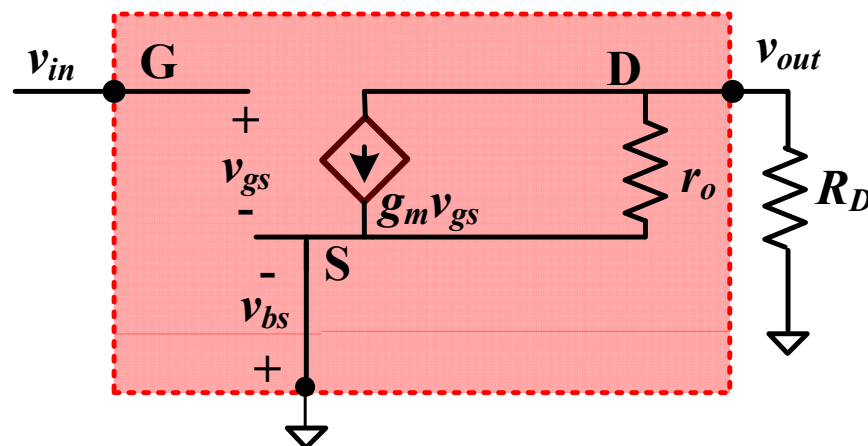
- **Method 2: Small-signal equivalent circuit**
 - **Case 1: Ideal MOS**



$$\left. \begin{aligned} v_{out} &= -i_d R_D \\ i_d &= g_m v_{in} \end{aligned} \right\} \longrightarrow A_v = \frac{v_{out}}{v_{in}} = -g_m R_D$$



Case 2: Considering channel length modulation



$$\left. \begin{aligned} v_{out} &= -i_d (R_D \parallel r_o) \\ i_d &= g_m v_{in} \end{aligned} \right\} \longrightarrow \begin{aligned} A_v &= -g_m \cdot (R_D \parallel r_o) \\ &\approx -g_m \cdot R_D \end{aligned}$$

-
- The maximum gain of a CS stage with resistive load

$$A_v = -g_m \cdot (R_D \parallel r_o)$$

$$A_{v,\max} = \lim_{R_D \rightarrow \infty} A_v = -g_m \cdot r_o$$

- This is called the **intrinsic gain** of CS stage
- Can we obtain this high gain with resistive load even though a infinitely large resistor can be realized?

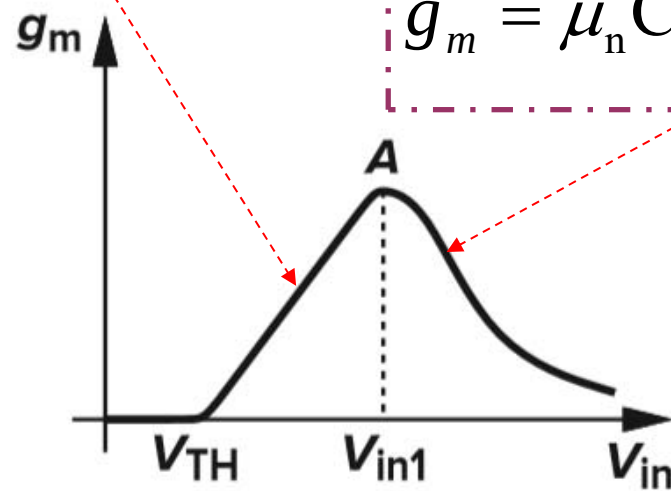
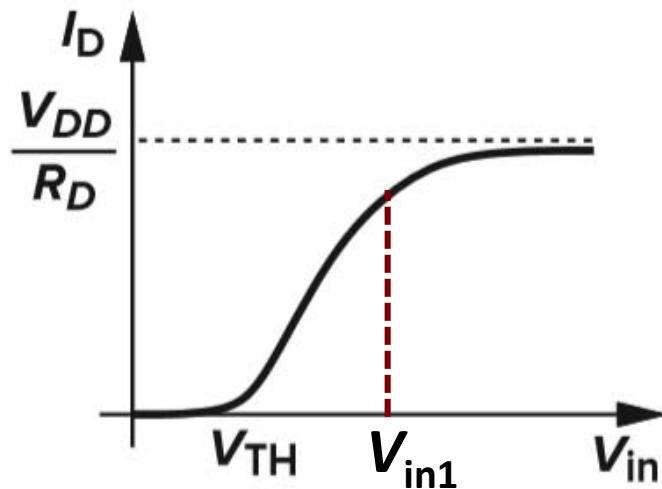
g_m as a function of V_{in}

**Saturation
region**

$$g_m = \mu_n C_{ox} \frac{W}{L} (V_{in} - V_{TH})$$

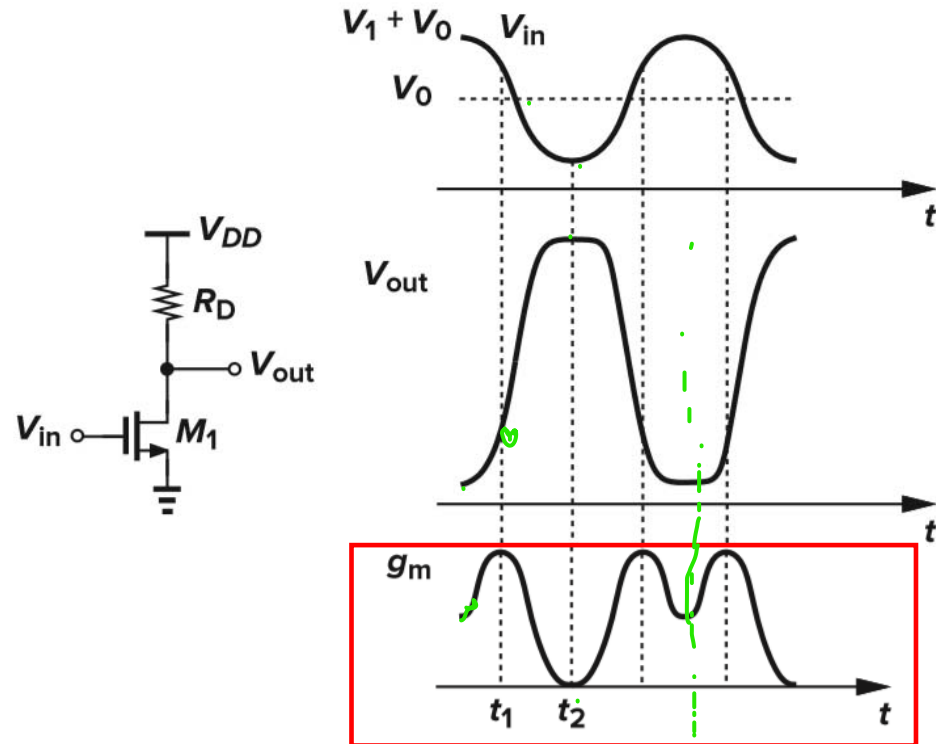
**Triode
region**

$$g_m = \mu_n C_{ox} \frac{W}{L} V_{DS}$$



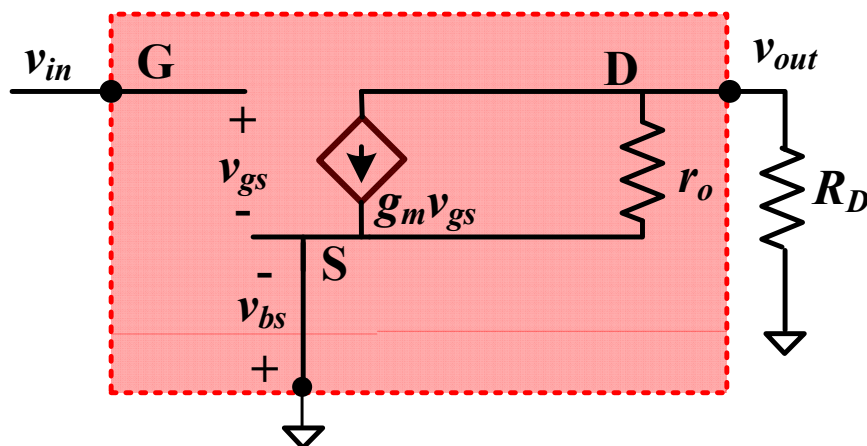
Example 3.2

- A CS driven by a sinusoid $V_{in} = V_1 \cos \omega_1 t + V_0$
- V_0 is the bias value
- V_1 is large enough to drive M1 into the off and triode regions.
- Sketch the g_m of the transistor as a function of time.



Input and output resistance

- Thevenin equivalent input and output resistance



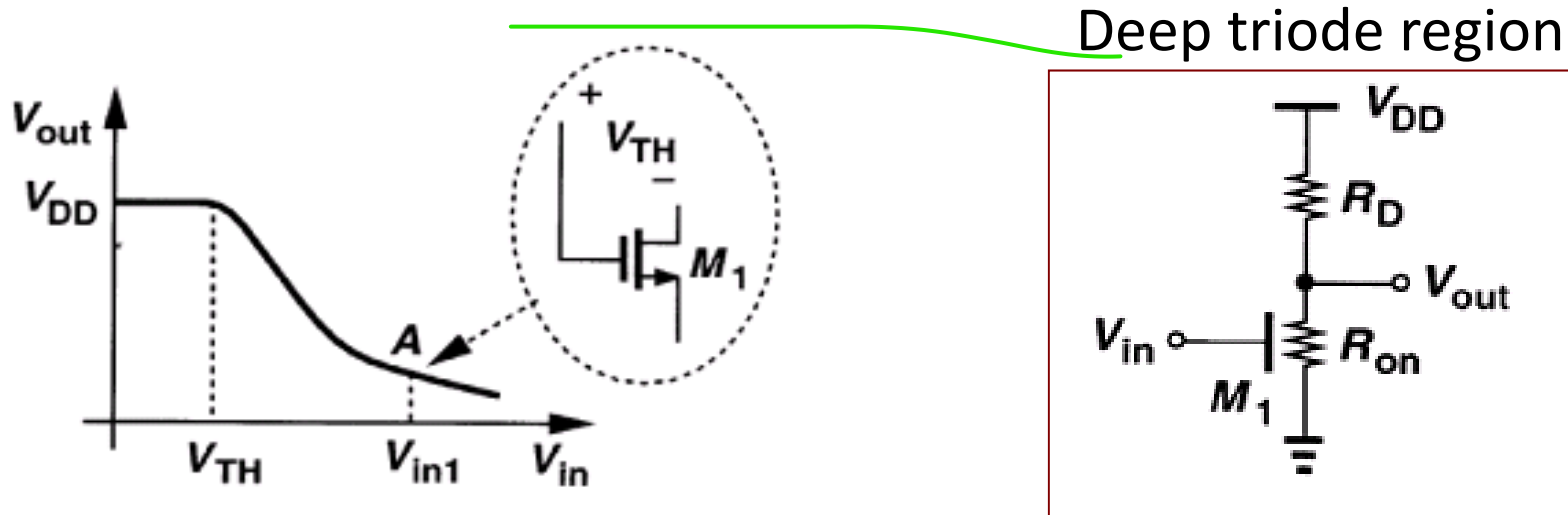
- Obtain input resistance with output shorted and output resistance with input shorted

$$r_{in} = \frac{V_{in}}{i_{in}} = \infty$$

$$r_{out} = r_o \parallel R_D \approx R_D$$

Behavior in triode range

- When $V_{in} > V_{in1}$, M_1 enters triode region.



- The slope of the input/output curve drops dramatically
 - Amplifier loses amplifying ability
 - If V_{in} is large enough, M_1 acts as a voltage controlled resistor



-
- How to calculate the exact value of V_{in1}

- $V_{DS} = V_{in1} - V_{TH}$

$$V_{in1} - V_{TH} = V_{DD} - \frac{\mu_n C_{ox}}{2} \frac{W}{L} (V_{in1} - V_{TH})^2 \cdot R_D$$

→ V_{in1} is a function of R_D .

- Larger R_D results in smaller V_{in1}

- Dilemma: trade off between gain and input range

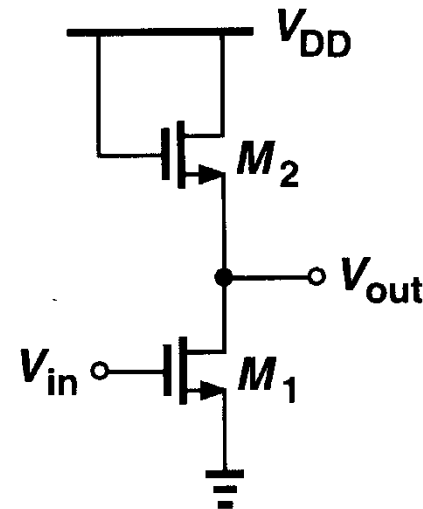
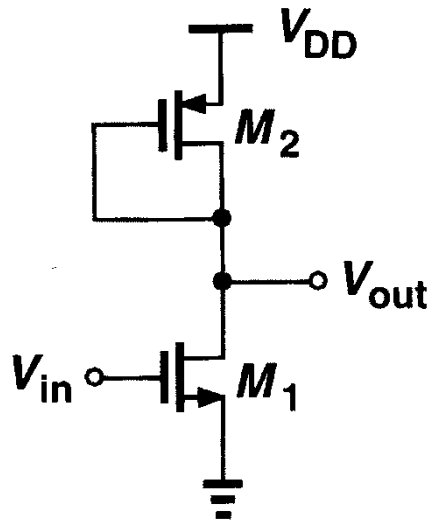
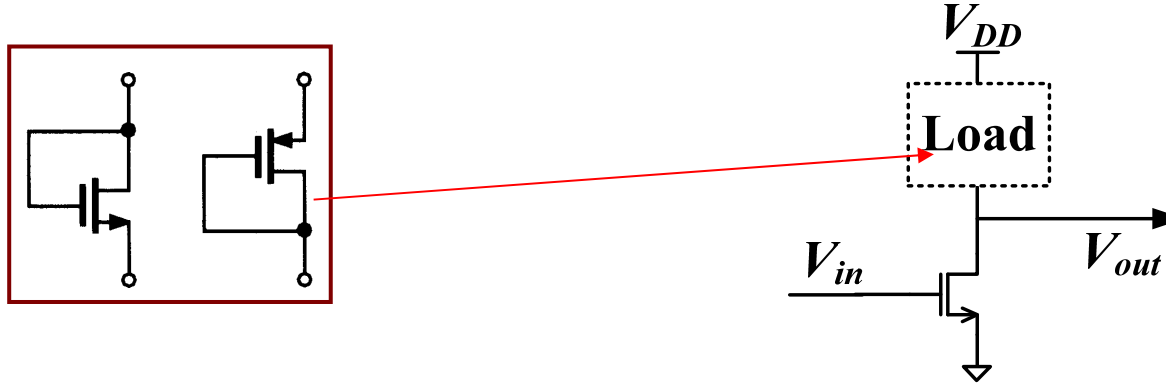
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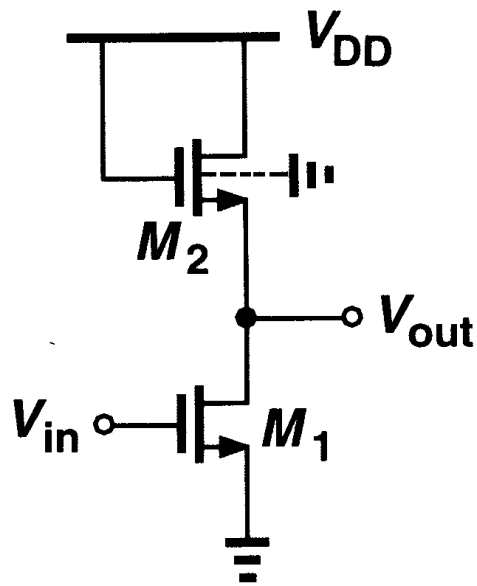
CS stage with diode-connected load

- Put a diode-connected transistor into the load box



Large-signal analysis

- Drain currents of M_1 and M_2 are the same: $I_{d1}=I_{d2}$



$$\frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right)_1 (V_{in} - V_{TH1})^2$$

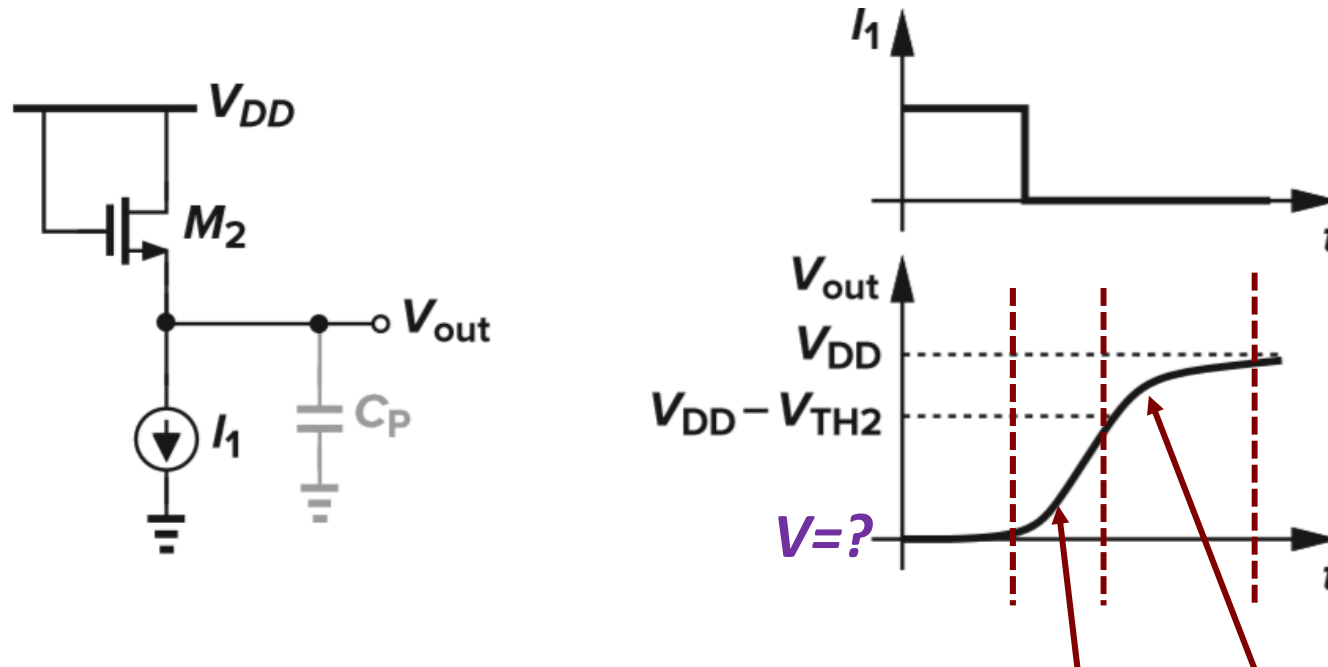
$$= \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right)_2 (V_{DD} - V_{out} - V_{TH2})^2$$

$$\sqrt{\left(\frac{W}{L} \right)_1} (V_{in} - V_{TH1}) = \sqrt{\left(\frac{W}{L} \right)_2} (V_{DD} - V_{out} - V_{TH2})$$

- V_{out} is almost a linear function of V_{in}



Large-signal output range

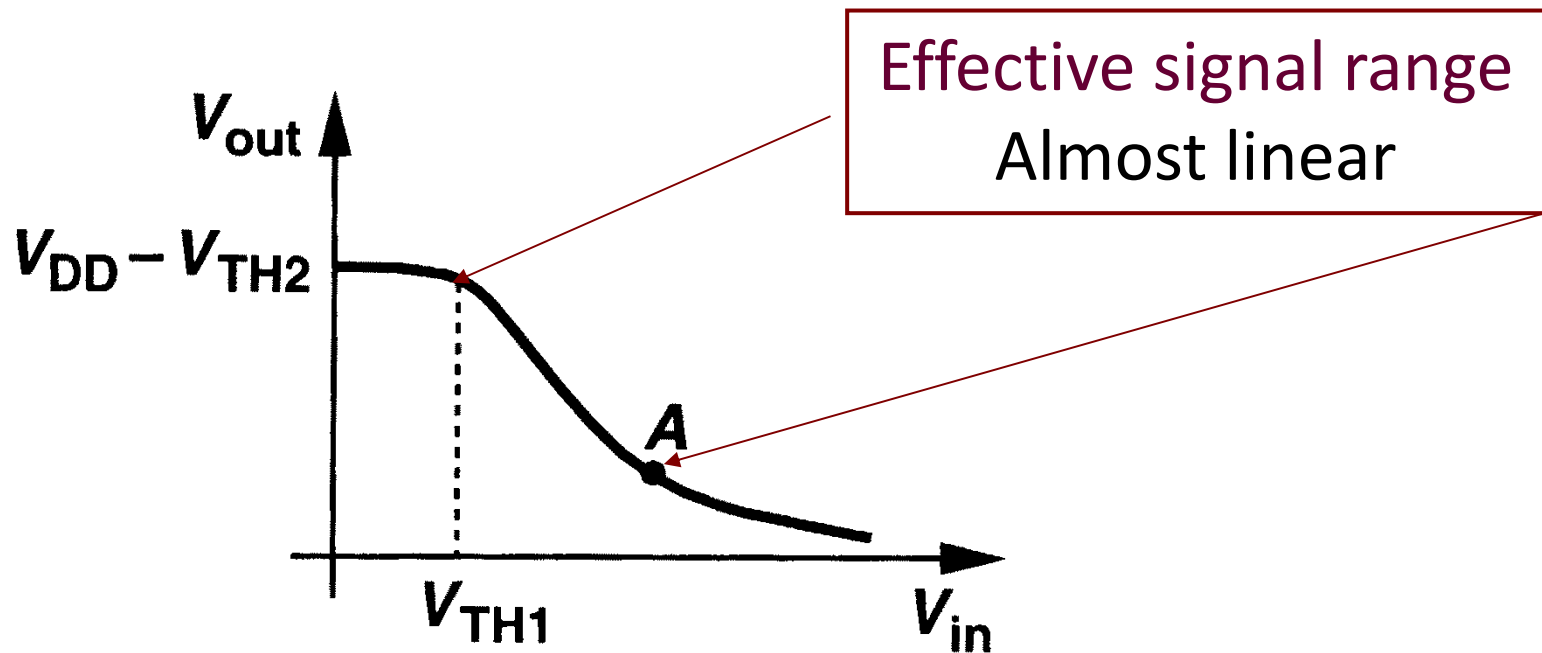


How to explain these 2 regions with different slope?

- **M2 is driven by a current step**
 - After I_1 changes to 0, V_{out} increases to $V_{DD} - V_{TH2}$ at first,
 - Rises slowly to V_{DD} because of the sub- V_{TH} current



Output Swing



- Ignoring the sub- V_{TH} current, maximum output voltage is $V_{DD} - V_{TH2}$
- In the region at right of point A, M_1 goes into triode region
 - Where does the non-linear behavior come from?



Small-signal gain

- **Method 1: calculate from the large-signal input/output characteristic**

$$\sqrt{\left(\frac{W}{L}\right)_1} (V_{in} - V_{TH1}) = \sqrt{\left(\frac{W}{L}\right)_2} (V_{DD} - V_{out} - V_{TH2})$$

$$\sqrt{\left(\frac{W}{L}\right)_1} = \sqrt{\left(\frac{W}{L}\right)_2} \left(-\frac{\partial V_{out}}{\partial V_{in}} - \frac{\partial V_{TH2}}{\partial V_{in}} \right)$$

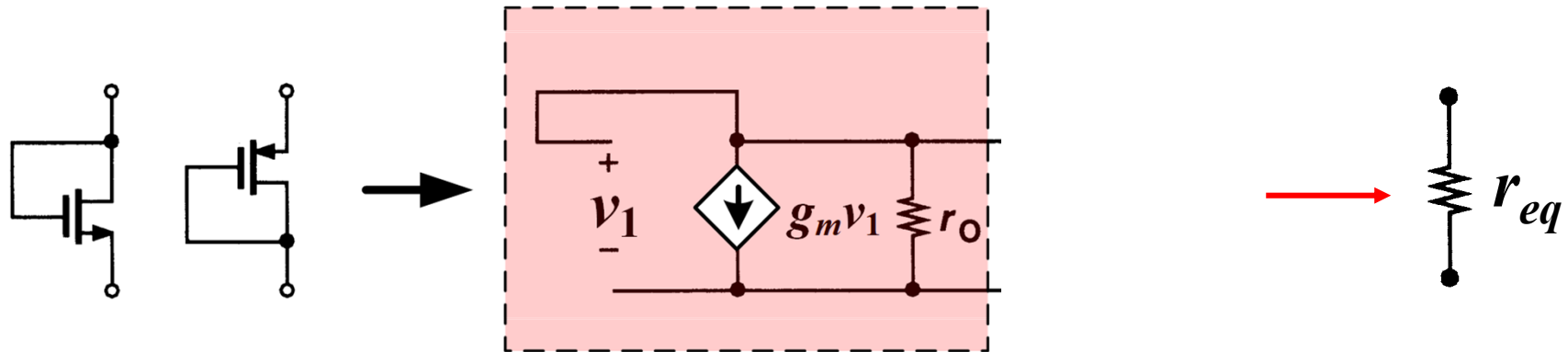
Body effect of M_2

$$\frac{\partial V_{TH2}}{\partial V_{in}} = \frac{\partial V_{TH2}}{\partial V_{out}} \cdot \frac{\partial V_{out}}{\partial V_{in}} = \eta \cdot \frac{\partial V_{out}}{\partial V_{in}}$$

$$A_v = \frac{\partial V_{out}}{\partial V_{in}} = -\sqrt{\frac{(W/L)_1}{(W/L)_2}} \cdot \frac{1}{1 + \eta}$$



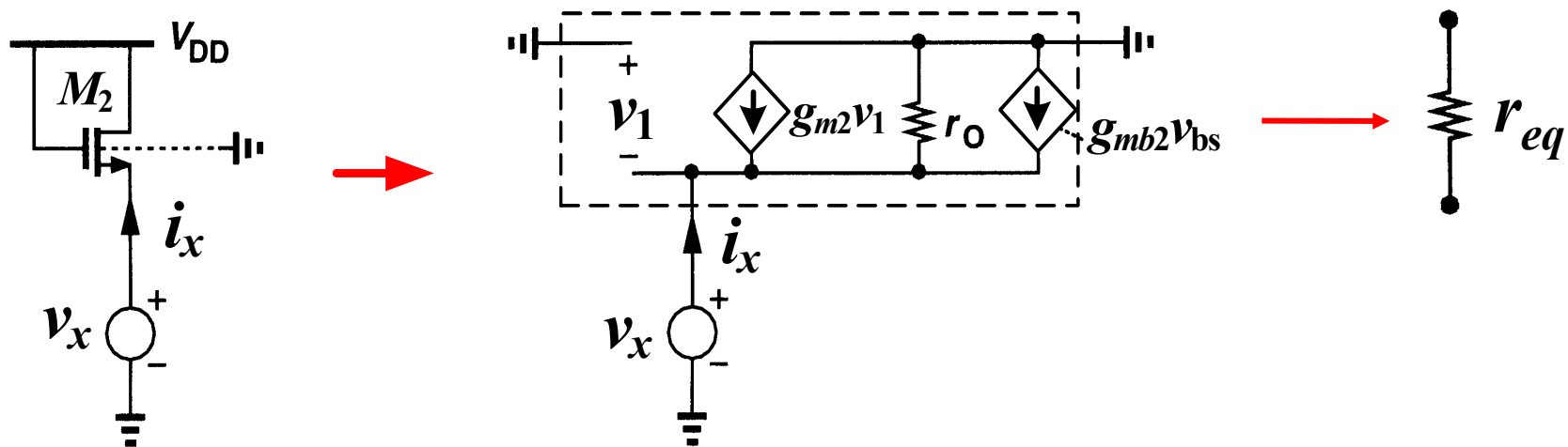
- Method 2: **small-signal equivalent circuit**
 - The small-signal model of diode-connect NMOS
 - It's a 2-terminal device (neglect body effect)



$$\left. \begin{aligned} i_x &= v_x / r_o + g_m v_1 \\ v_1 &= v_x \end{aligned} \right\} \rightarrow r_{eq} = \frac{v_x}{i_x} = r_o \parallel \frac{1}{g_m} \approx \frac{1}{g_m}$$



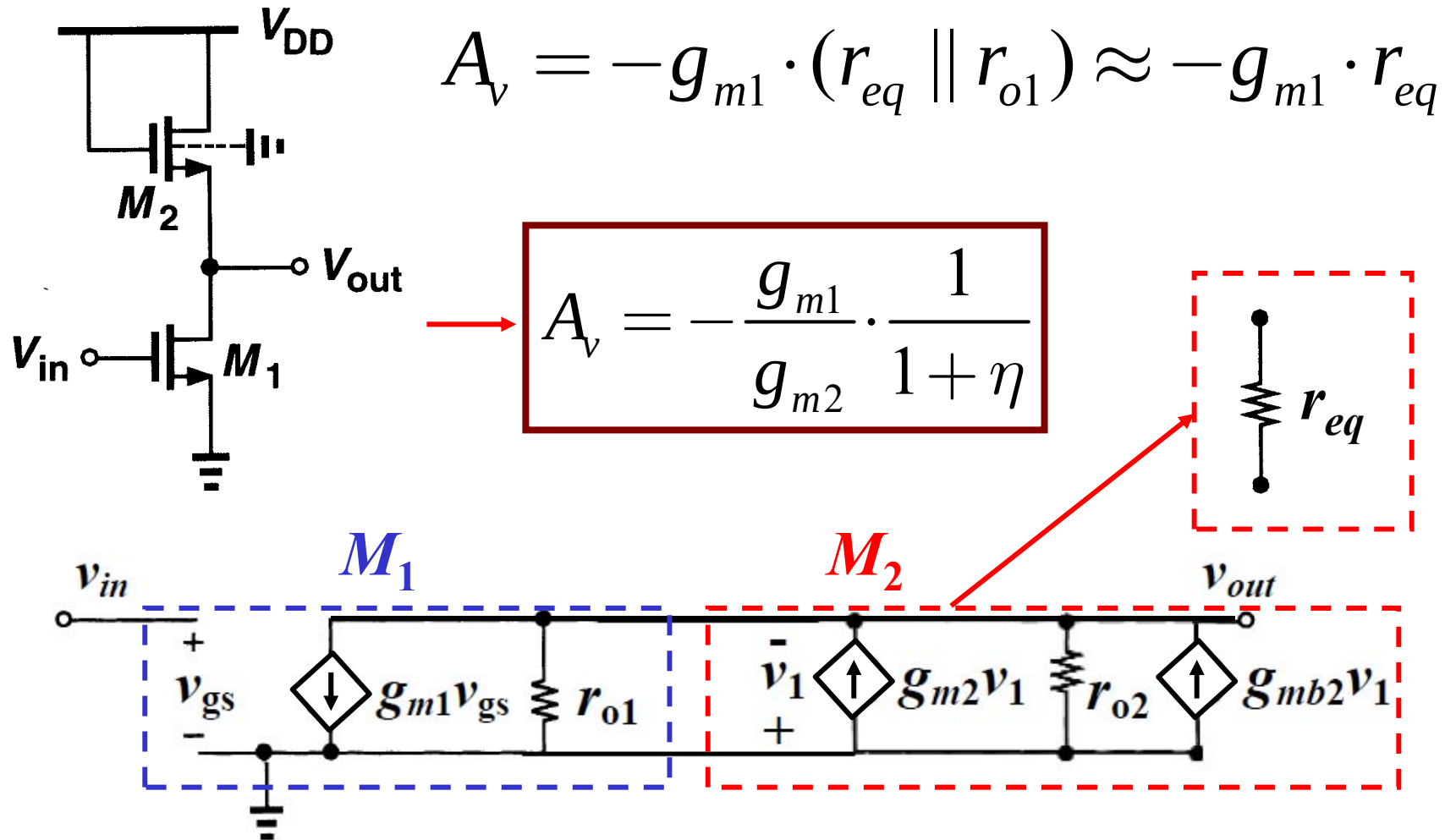
- Small signal resistance considering body effect



$$i_x = \frac{v_x}{r_o} + (g_{m2} + g_{mb2})v_x$$

$$\rightarrow r_{eq} = \frac{v_x}{i_x} = r_o \parallel \frac{1}{g_{m2} + g_{mb2}} \approx \frac{1}{g_{m2} + g_{mb2}} = \frac{1}{(1 + \eta)g_{m2}}$$

- Calculate A_v with small signal equivalent circuit



$$A_v = -\frac{g_{m1}}{g_{m2}} \cdot \frac{1}{1 + \eta}$$

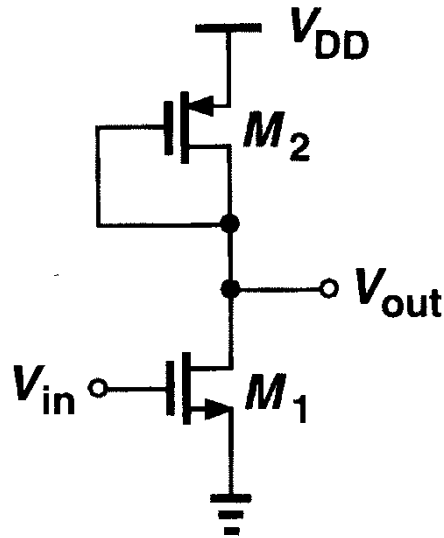
$$A_v = -\frac{\sqrt{2\mu_n C_{ox} (W/L)_1 I_{D1}}}{\sqrt{2\mu_n C_{ox} (W/L)_2 I_{D2}}} \cdot \frac{1}{1 + \eta}$$

$$= -\sqrt{\frac{(W/L)_1}{(W/L)_2}} \cdot \frac{1}{1 + \eta}$$

- Same results as that of large-signal analysis
- A_v is independent of biasing current, a linear amplifier, but is still influenced by body effect



Use PMOS load to avoid body effect



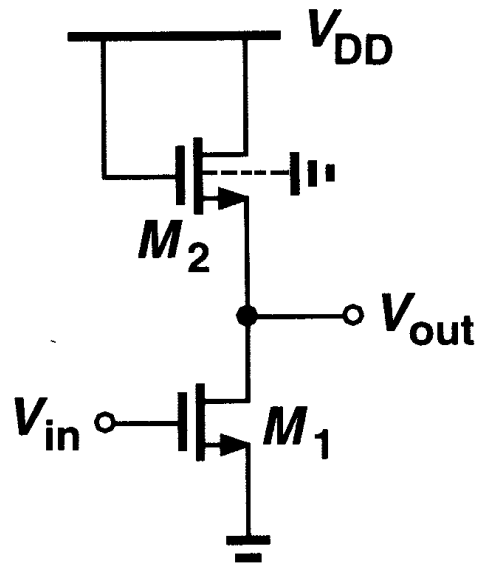
$$A_v = -\frac{g_{m1}}{g_{m2}}$$

$$A_v = -\sqrt{\frac{\mu_n (W/L)_1}{\mu_p (W/L)_2}}$$

– No body effect, more linear than NMOS load



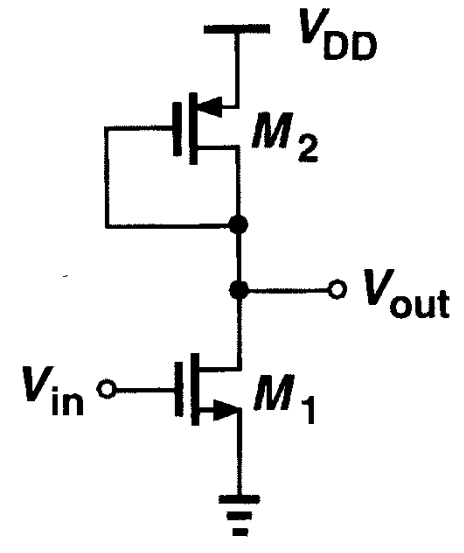
Small-signal input and output resistance



$r_{in} = \infty$

$$r_{out} = r_{o1} \parallel r_{o2} \parallel \frac{1}{g_{m2}(1+\eta)}$$

$$\approx \frac{1}{g_{m2}(1+\eta)}$$



$r_{in} = \infty$

$$r_{out} = r_{o1} \parallel r_{o2} \parallel \frac{1}{g_{m2}}$$

$$\approx \frac{1}{g_{m2}}$$

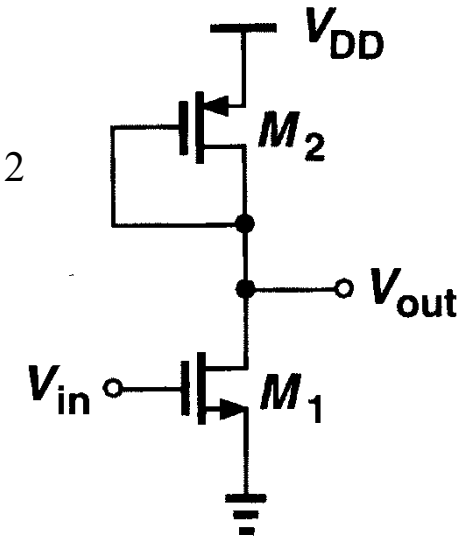


Signal range

- Because $I_{D1}=I_{D2}$, then

$$u_n \left(\frac{W}{L} \right)_1 (V_{GS1} - V_{TH1})^2 = u_p \left(\frac{W}{L} \right)_2 (V_{GS2} - V_{TH2})^2$$

$$\rightarrow \frac{|V_{GS2} - V_{TH2}|}{(V_{GS1} - V_{TH1})} = \sqrt{\frac{u_n (W/L)_1}{u_p (W/L)_2}} = |A_v|$$

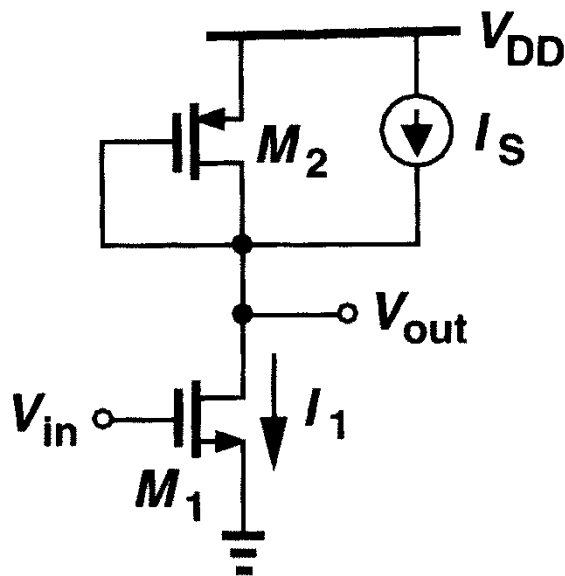


- If $V_{GS1} - V_{TH1} = 0.2V$, $|V_{TH2}| = 0.7V$, and we want $A_v = 10$, then $V_{GS2} = 2.7V$! (considering $V_{DD} = 3V$)

- Hard trade-off between signal range and gain

Example 3.6

- M_1 is biased in saturation region, $I_S = 0.75 I_1$
 - Calculate A_v
 - Analyze the trade-off between A_v and signal range



$$A_v = -\frac{g_{m1}}{g_{m2}} = -\sqrt{\frac{4\mu_n C_{ox} (W/L)_1}{\mu_p C_{ox} (W/L)_2}}$$

Gain doubles, while power consumption keeps the same!



- **Signal range**

$$\mu_n \left(\frac{W}{L} \right)_1 (V_{GS1} - V_{TH1})^2 = 4 \cdot \mu_p \left(\frac{W}{L} \right)_2 (V_{GS2} - V_{TH2})^2$$

$$\frac{|V_{GS2} - V_{TH2}|}{(V_{GS1} - V_{TH1})} = \sqrt{\frac{\mu_n (W/L)_1}{4\mu_p (W/L)_2}} = \frac{|A_v|}{4}$$

– Under the former conditions, $|V_{GS2}|$ reduces to 1.2V, output range is enlarged.

- **This is a very effective circuit.**

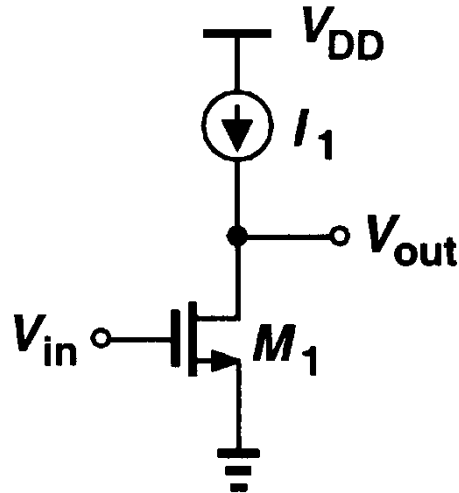
Please think over Example 3.7 by yourself.

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CS stage with current source load



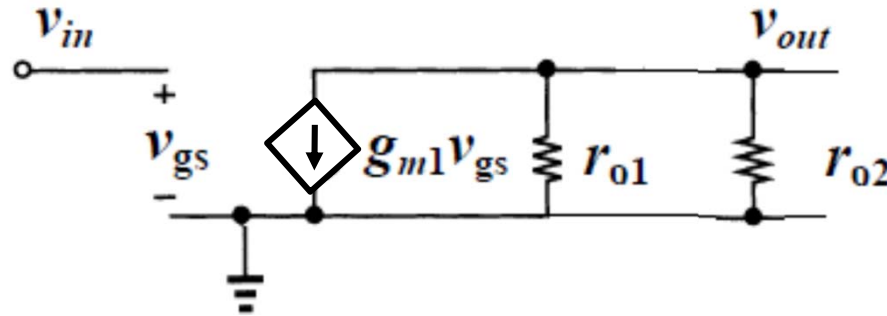
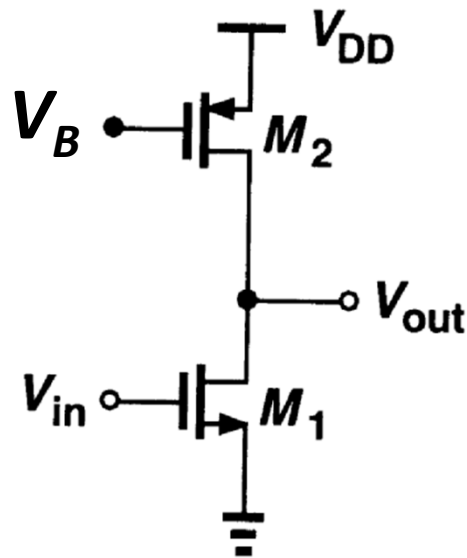
- The small signal resistance for an ideal current source is ∞ , then small signal gain is

$$A_v = -g_m \cdot r_o \quad \text{The intrinsic gain}$$



CS stage with current source load

- The current source load is realized with a PMOS
- Use small-signal equivalent circuit
 - How to model M_2 ?

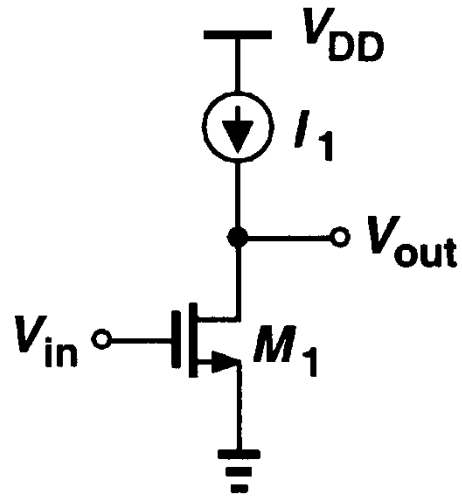


$$A_v = -g_m \cdot (r_{o1} \parallel r_{o2})$$

$$r_{in} = \infty \quad r_{out} = r_{o1} \parallel r_{o2}$$



Current source load v.s. resistive load

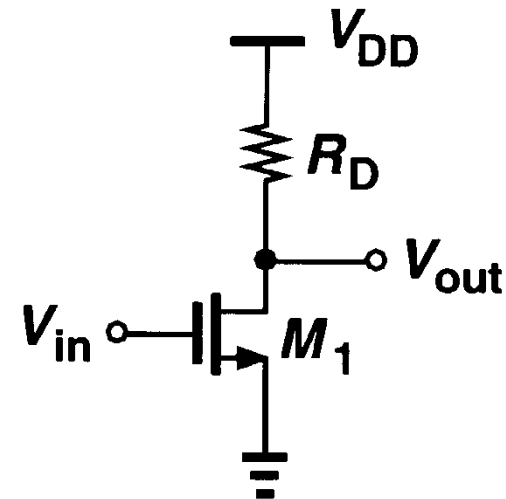


$$A_v = -g_m \cdot r_o$$

$$= -\sqrt{2\mu_n C_{ox} \frac{W}{L} I_D} \cdot \frac{1}{\lambda I_D}$$

$$\rightarrow A_v \propto 1/\sqrt{I_D}$$

Why??



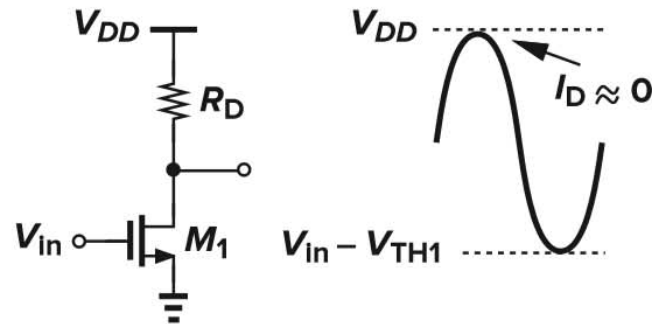
$$A_v = -g_m \cdot R_D$$

$$= -\sqrt{2\mu_n C_{ox} \frac{W}{L} I_D} \cdot R_D$$

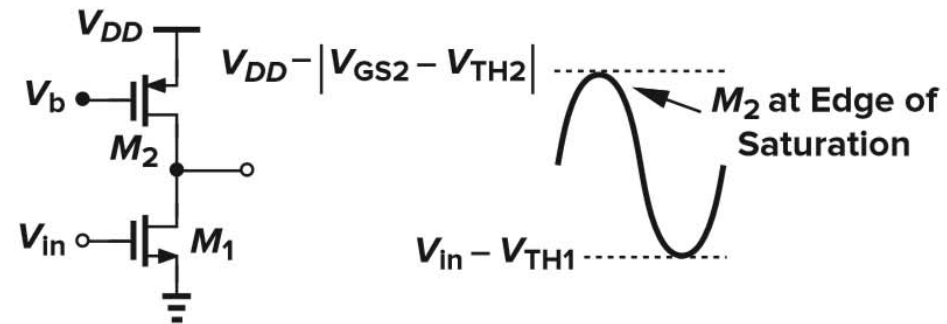
$$\rightarrow A_v \propto \sqrt{I_D}$$



Comparison of output swing



- Low gain
- High swing

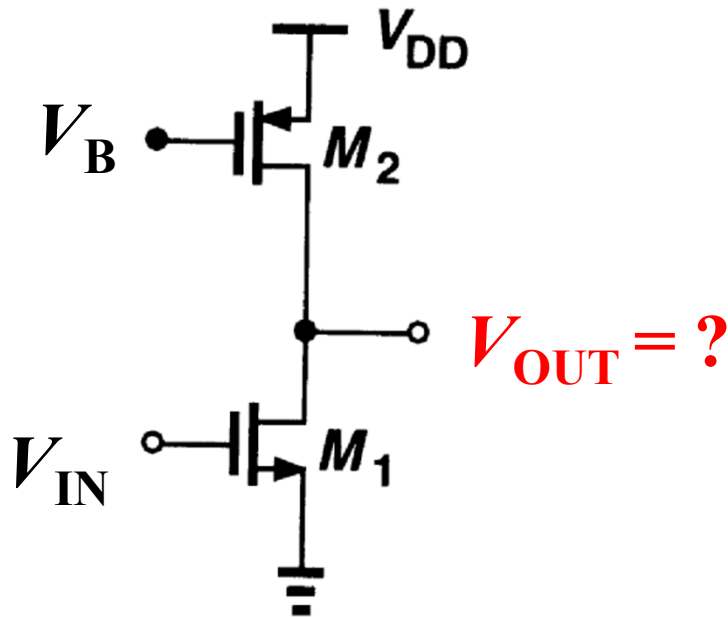


- High gain
- Lower swing



DC output voltage

- What is the DC output voltage?
 - Consider 2 cases: $\lambda=0$ and $\lambda\neq 0$



DC output voltage is very sensitive to PVT

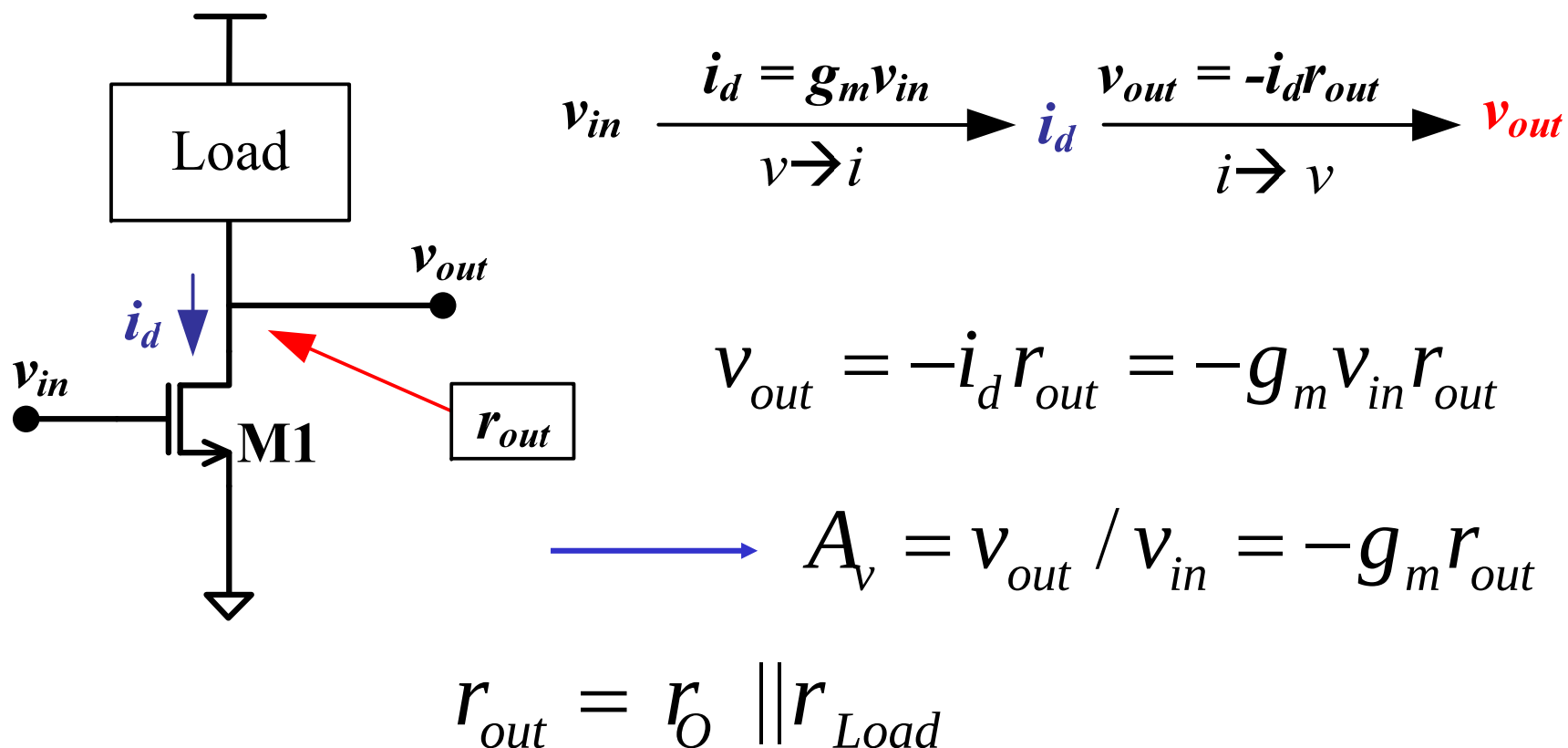


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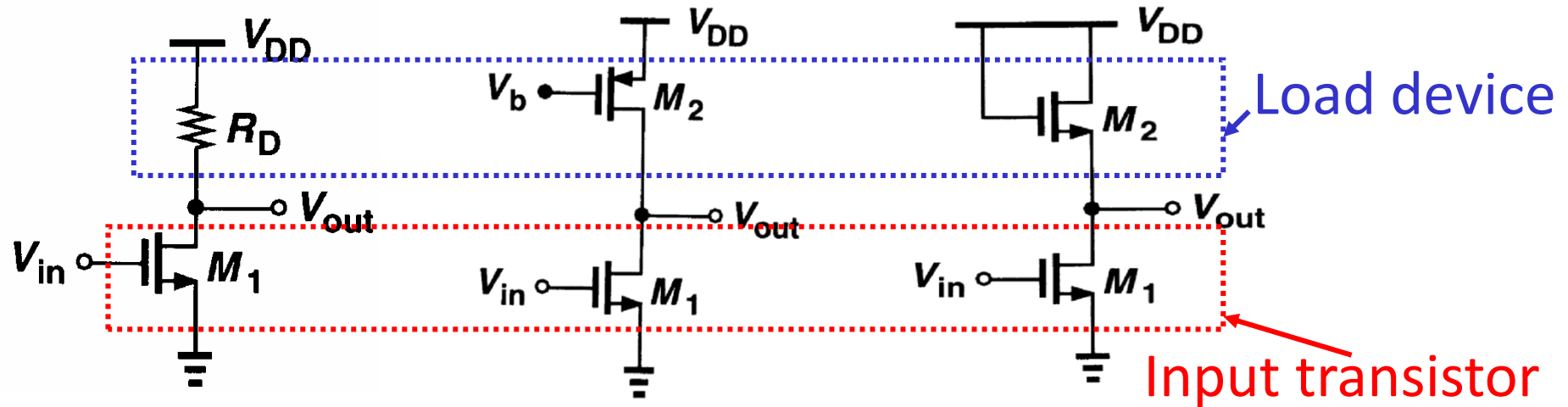
Versatile method for calculating A_v



Key point is to calculate output resistance !



Output resistance summary



$$r_{out} \approx R_D$$

Medium

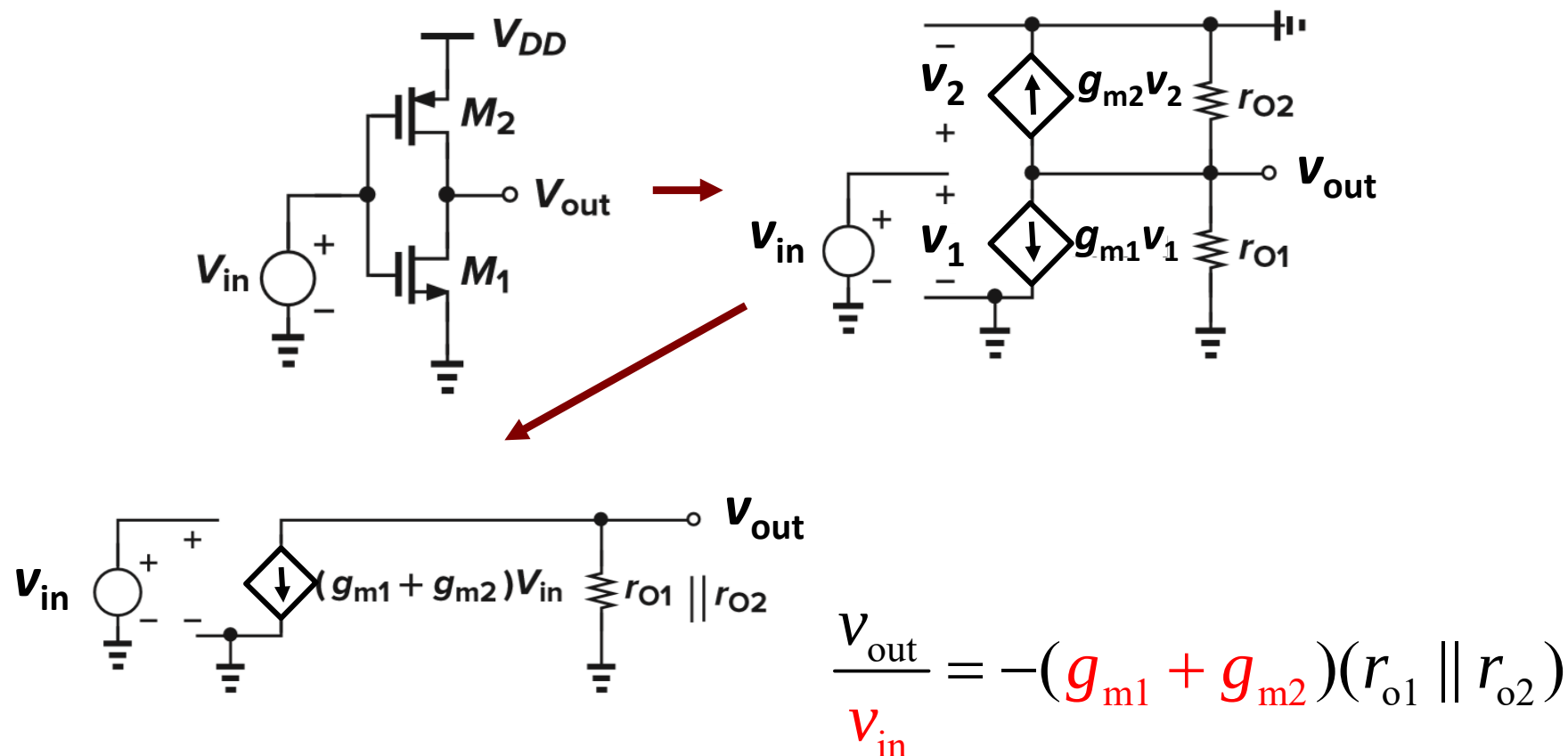
$$r_{out} = r_{o1} \parallel r_{o2}$$

High

$$r_{out} \approx \frac{1}{g_{m2}(1+\eta)}$$

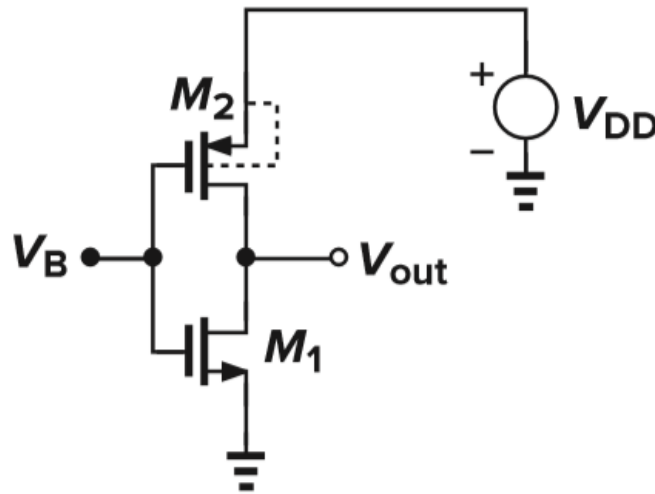
Low

Class-AB stage (CS stage with active load)



- Looks like an inverter
- Both M_1 and M_2 act as the CS input transistor
- A dc current provides $2 g_m \rightarrow$ **concept of current reuse**

Sensitivity to V_{DD} 's variation



$$\frac{V_{out}}{V_{dd}} = \frac{g_{m2} r_{o2} + 1}{r_{o1} + r_{o2}} r_{o1}$$

$$= \left(g_{m2} + \frac{1}{r_{o2}} \right) (r_{o1} \parallel r_{o2})$$

- The original Class-AB stage is very sensitive to V_{DD}
 - $V_{GS1} + V_{GS2} = V_{DD}$; Change in $V_{DD} \rightarrow$ change in I_D
- Needs complex bias circuit to solve it
 - Try to come up with a solution by yourself



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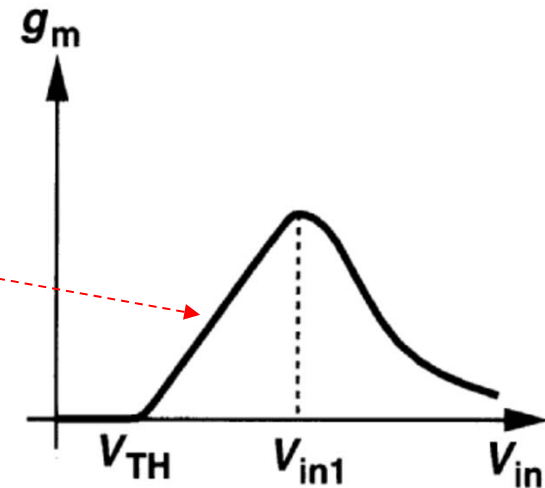
CS Stage with source degeneration

- Problem of CS stage:

- Nonlinearity! Where is the nonlinearity from?

$$A_v = -g_m R_D$$

$$g_m = \mu_n C_{ox} \frac{W}{L} (V_{in} - V_{TH})$$



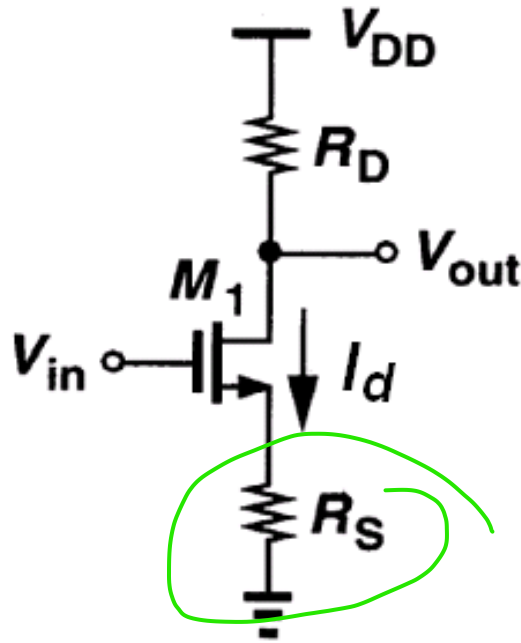
- The g_m is not constant but a function of the input signal! How can we deal with this ?

- Place a resistor in series with S of the MOS

- Source degeneration



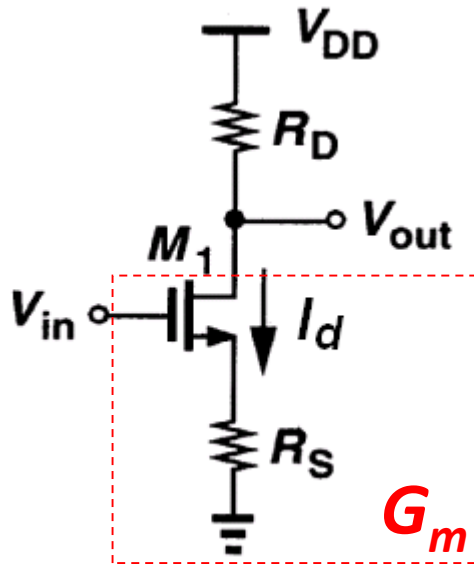
Source degeneration: observation



- When V_{in} increases:
 - I_d increases
 - Voltage drop on R_S also increases
 - A fraction of V_{in} drops on R_S
 - Leading to smooth variation in I_D
- What is the exact relationship between V_{in} and I_d ?



Large signal analysis



- Assuming $\lambda = \gamma = 0$, I_d can still be expressed as:

$$\begin{aligned} I_d &= \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{gs} - V_{TH})^2 \\ &= \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{in} - R_S I_d - V_{TH})^2 \end{aligned}$$

- V_{in} can still control I_d , but the control ability is not g_m now
- We define the **equivalent transconductance** as

$$G_m = \frac{\partial I_d}{\partial V_{in}}$$



Finding G_m and A_v from large-signal analysis

$$G_m = \frac{\partial I_d}{\partial V_{in}} = \underbrace{\mu_n C_{ox} \frac{W}{L}}_{g_m} (V_{in} - R_S I_d - V_{TH})(1 - R_S G_m)$$

$$G_m = g_m (1 - R_S G_m)$$

$$G_m = \frac{g_m}{1 + g_m R_S}$$

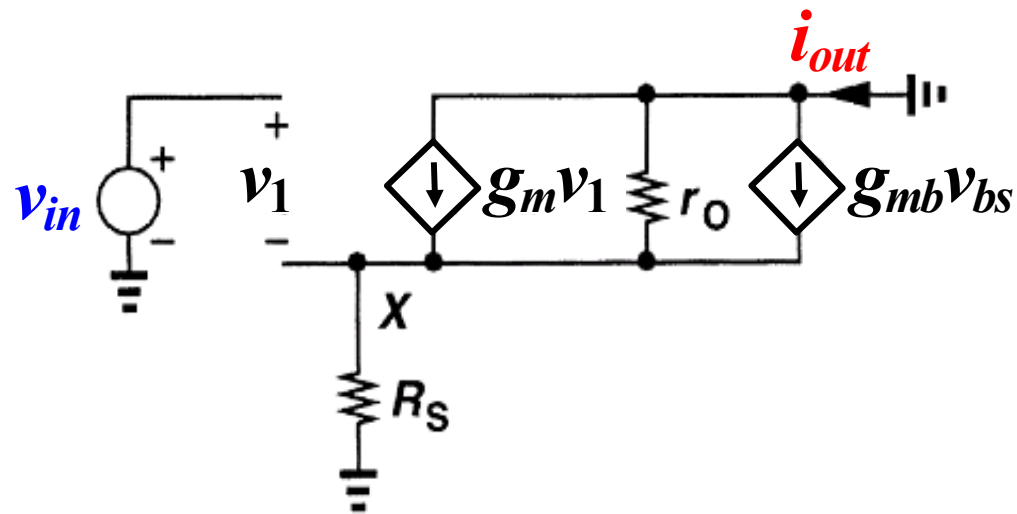
- Small-signal gain can be expressed directly as:

$$A_v = -G_m R_D = -\frac{g_m R_D}{1 + g_m R_S}$$

- If R_S is large enough $\rightarrow G_m \approx 1/R_S, A_v = R_D/R_S$
- High linearity is obtained, **at the price of lower gain**

Finding G_m from small-signal model

- Considering both body effect and channel length modulation:
- Connect output to ac ground to find output current



$$v_1 = v_{in} - v_x$$

$$v_x = -v_{bs} = R_S i_{out}$$

$$\begin{aligned}
 i_{out} &= g_m v_1 + g_{mb} v_{bs} + \frac{-v_x}{r_o} \\
 &= g_m (v_{in} - R_S i_{out}) + g_{mb} (-R_S i_{out}) + \left(-\frac{R_S i_{out}}{r_o}\right)
 \end{aligned}$$

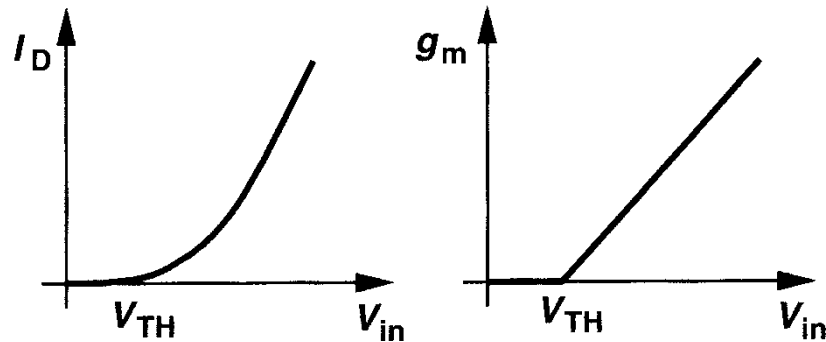
→

$$G_m = \frac{i_{out}}{v_{in}} = \frac{g_m}{1 + g_m R_S + g_{mb} R_S + R_S / r_o}$$

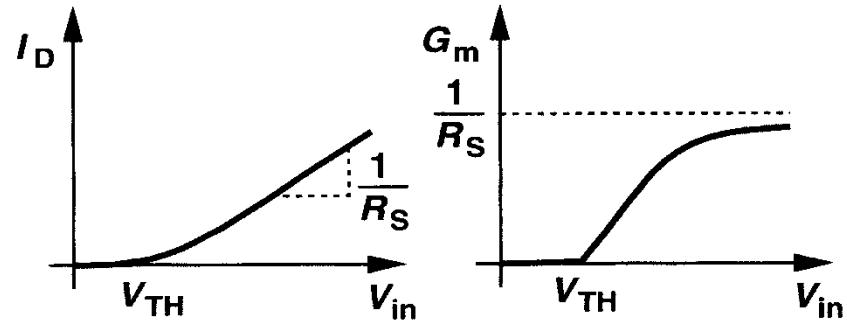
- If $\gamma=0$ ($g_{mb}=0$), and $\lambda=0$ ($r_o=\infty$), the results is the same as large-signal analysis



The effect of source degeneration



CS without degeneration



CS with degeneration

- When V_{in} is very small, both cases are almost the same
 - $1/g_m \gg R_S, \rightarrow G_m \approx g_m$
- When V_{in} increases, degeneration works
 - I_D is almost linear, G_m approaches $1/R_S$
- Trade gain for linearity!!



Output resistance: inspection

- When the drain voltage increases by ΔV

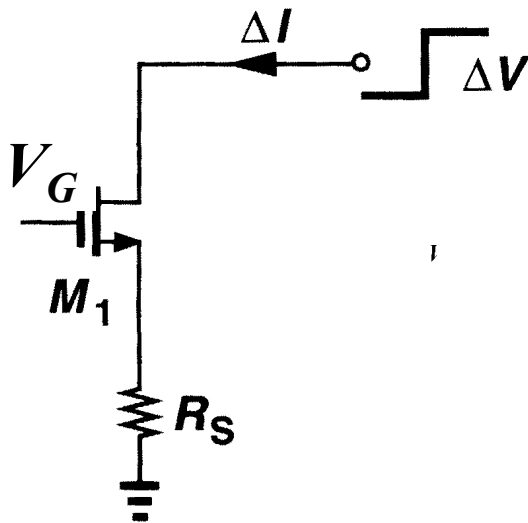
- I_D increases a bit because of channel length modulation effect of M_1

- Voltage drop on R_S increases

- V_{GS} of M_1 decreases

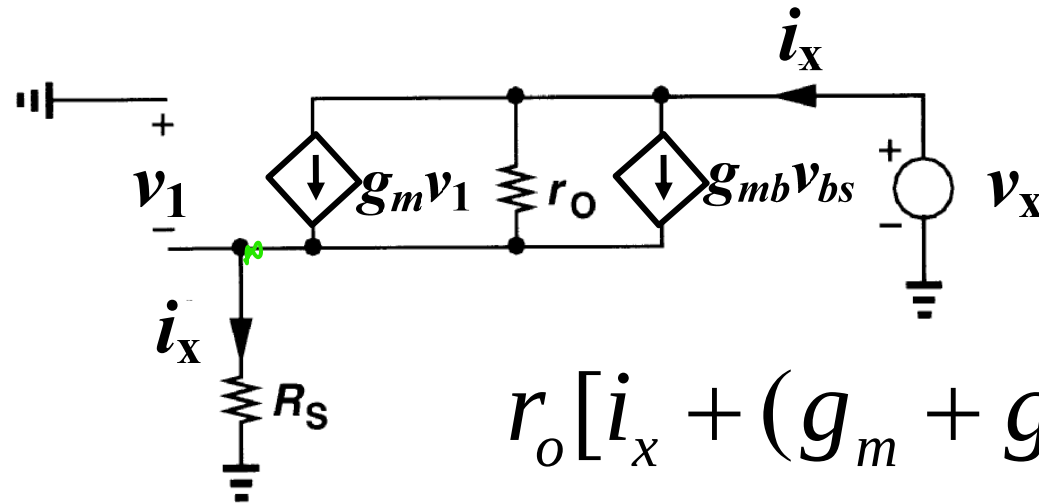
- The feedback keeps I_D almost constant

- **Output resistance increases!**



Output Resistance

- Consider only M_1 and R_S



The current flowing through r_o

$$i_x - (g_m + g_{mb})v_1$$

$$v_1 = -i_x R_S$$

$$r_o [i_x + (g_m + g_{mb})i_x R_S] + i_x R_S = v_x$$

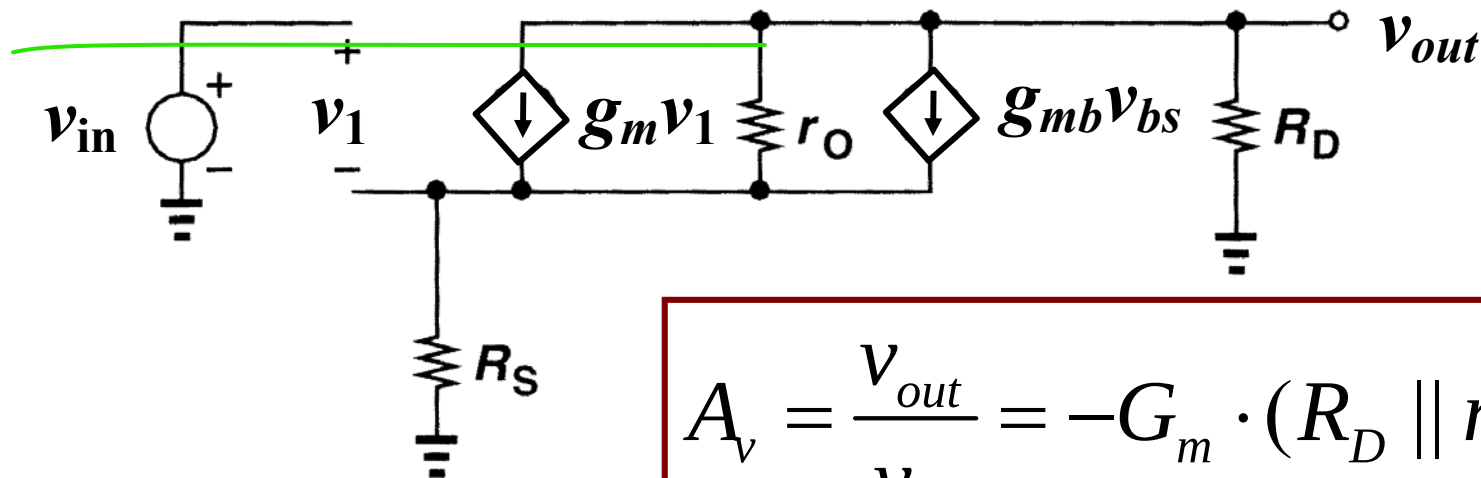
$$r_{OD} = v_x / i_x = R_S + r_o + (g_m + g_{mb})r_o \cdot R_S$$

$$\approx (g_m + g_{mb})r_o \cdot R_S$$

The output resistance is increased by degeneration!!

Complete small-signal equivalent circuit

- Calculate the small signal gain based on above analysis
 - First, v_{in} is converted i_d into current by G_m
 - Then i_d flows into the parallel combination of r_{od} and R_D to generate v_{out}



$$A_v = \frac{v_{out}}{v_{in}} = -G_m \cdot (R_D \parallel r_{OD})$$



$$\begin{aligned}
 A_v &= \frac{V_{out}}{V_{in}} = -G_m \cdot (r_{OD} \parallel R_D) \\
 &= -\frac{g_m}{1 + g_m R_S + g_{mb} R_S + R_S / r_o} \cdot \{ [R_S + r_O + (g_m + g_{mb}) r_O R_S] \parallel R_D \}
 \end{aligned}$$

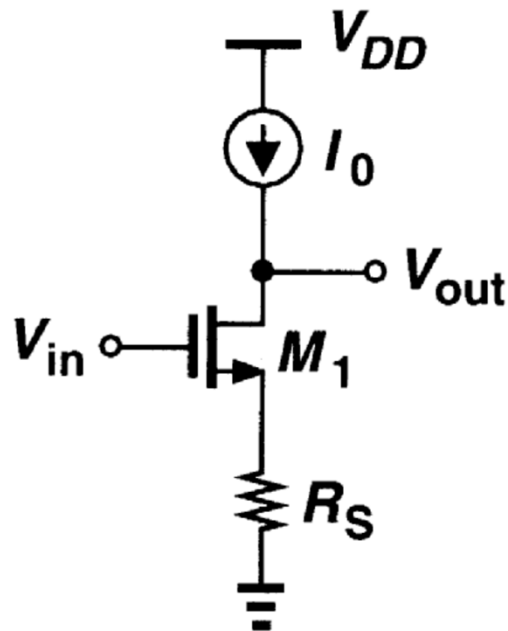
- **Total output resistance is**

$$r_{out} = R_D \parallel r_{OD} \approx R_D$$

Example 3.11

- Calculate small-signal gain

$$A_v = -G_m (R_D \parallel r_{OD})$$



$$G_m = \frac{g_m}{1 + R_S / r_O + (g_m + g_{mb}) R_S}$$
$$= \frac{g_m r_O}{r_O + [1 + (g_m + g_{mb}) r_O] R_S} = \frac{g_m r_O}{r_{OD}}$$

$$R_D \parallel r_{OD} = \infty \parallel r_{OD} = r_{OD}$$

$$A_v = -G_m R_{OUT} = -G_m r_{OD} = -g_m r_O \quad !$$



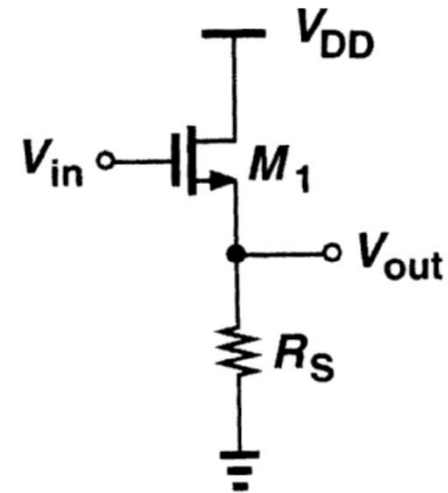
Outline

- Basic Concepts
- Common-Source stage
- **Source Follower**
- Common-Gate stage
- Cascode Stage



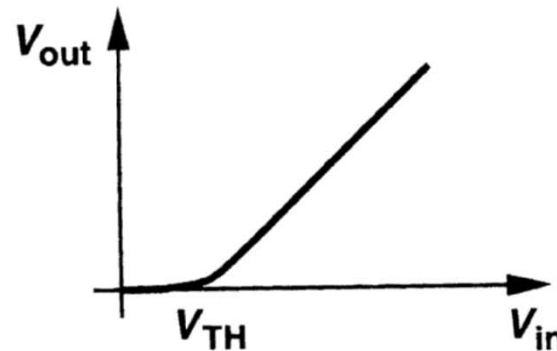
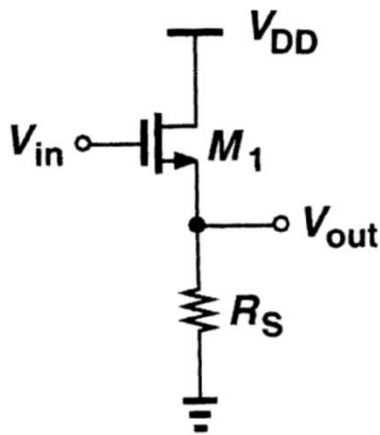
Source follower as a voltage buffer

- For a CS stage, the output resistance must be large to obtain high gain
 - Can't drive small load resistance
 - A voltage buffer is needed!
- Source follower
 - Input at gate, output at source
 - “common drain” stage



Large-signal analysis

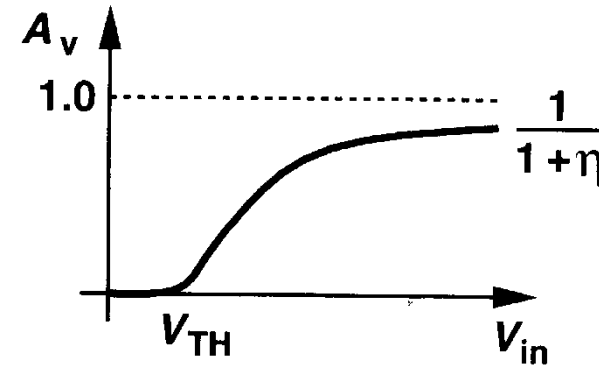
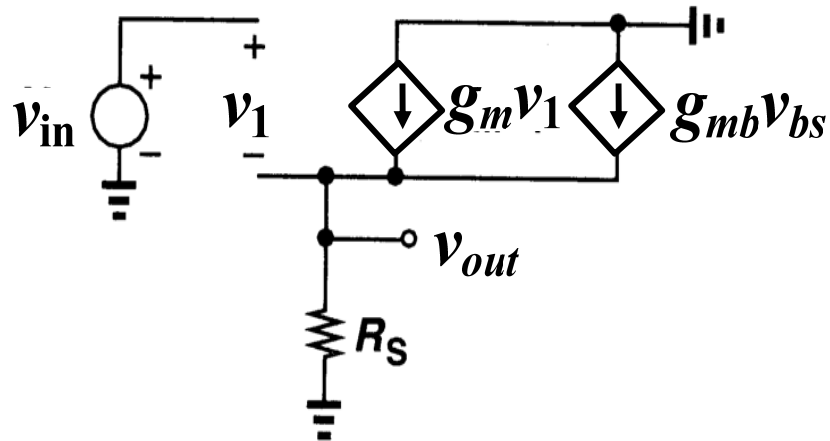
- When $V_{in} < V_{TH}$, M_1 is off and $V_{out} = 0$
- As $V_{in} > V_{TH}$, M_1 is on, I_{D1} flows through R_S
- When V_{in} increases further, V_{out} follows V_{in} with a difference equal to V_{GS}



$$\frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{in} - V_{TH} - V_{out})^2 R_S = V_{out} \longrightarrow A_v = \frac{\partial V_{out}}{\partial V_{in}} = \dots$$



Small-signal circuit



$$V_{bs} = -V_{out} \quad g_m v_1 - g_{mb} v_{out} = v_{out} / R_S$$

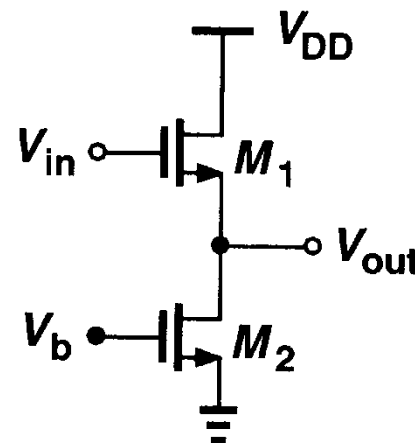
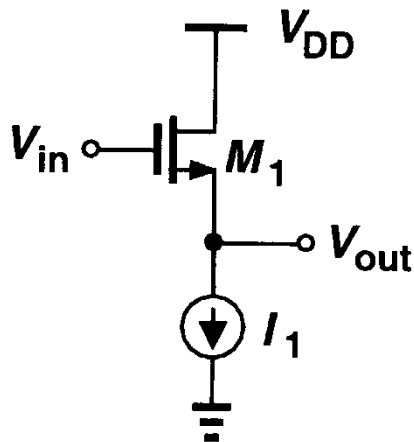
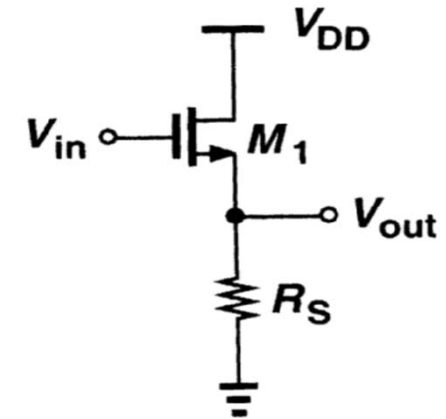
$$V_{in} - V_1 = V_{out}$$

$$\rightarrow A_v = \frac{v_{out}}{v_{in}} = \frac{g_m R_S}{1 + (g_m + g_{mb}) R_S}$$



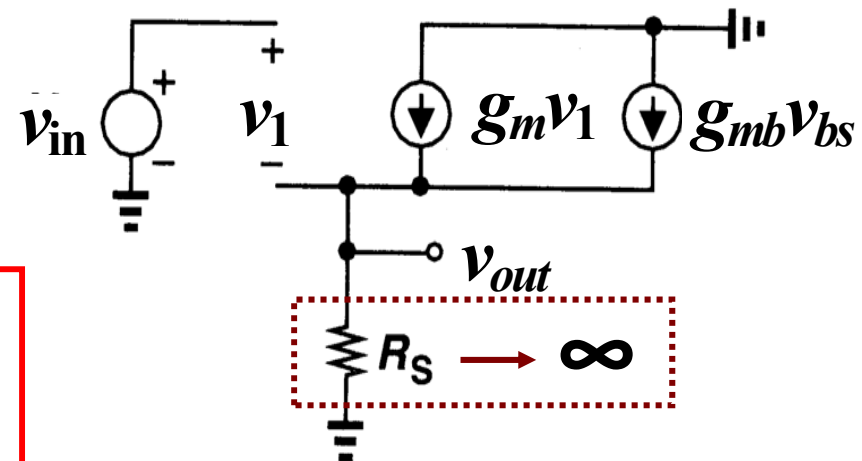
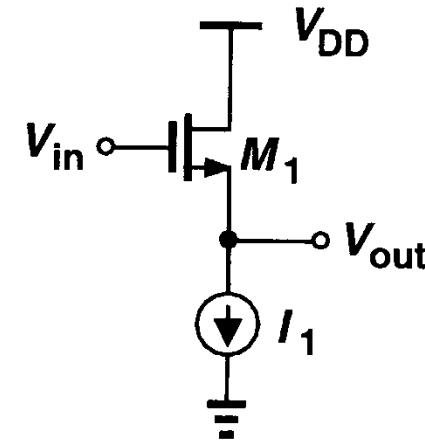
Replace the resistor with a current source

- I_D is heavily dependent on V_{in}
 - V_{GS} is not constant
 - Introduces large nonlinearity!
- Solution: replace the resistor with a current source



Obtain small-signal gain

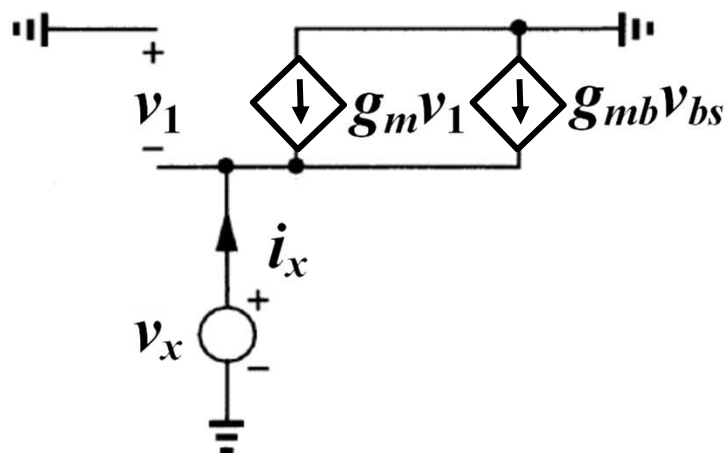
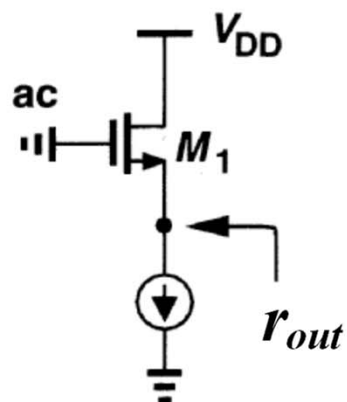
$$A_v = \frac{g_m R_S}{1 + (g_m + g_{mb}) R_S}$$
$$= \frac{g_m}{g_m + g_{mb}}$$



**A_v always smaller than 1
because of body effect!**



Output resistance



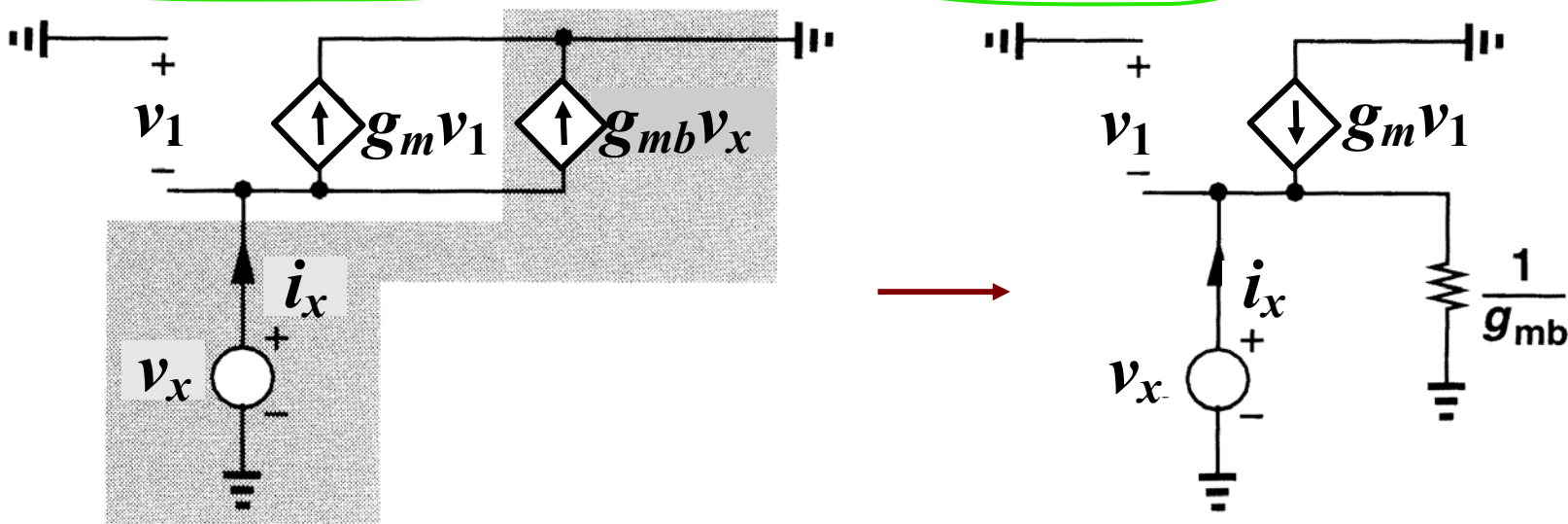
$$v_1 = v_{bs} = -v_x$$

$$i_x = g_m v_x + g_{mb} v_x \longrightarrow r_{out} = \frac{v_x}{i_x} = \frac{1}{g_m + g_{mb}}$$

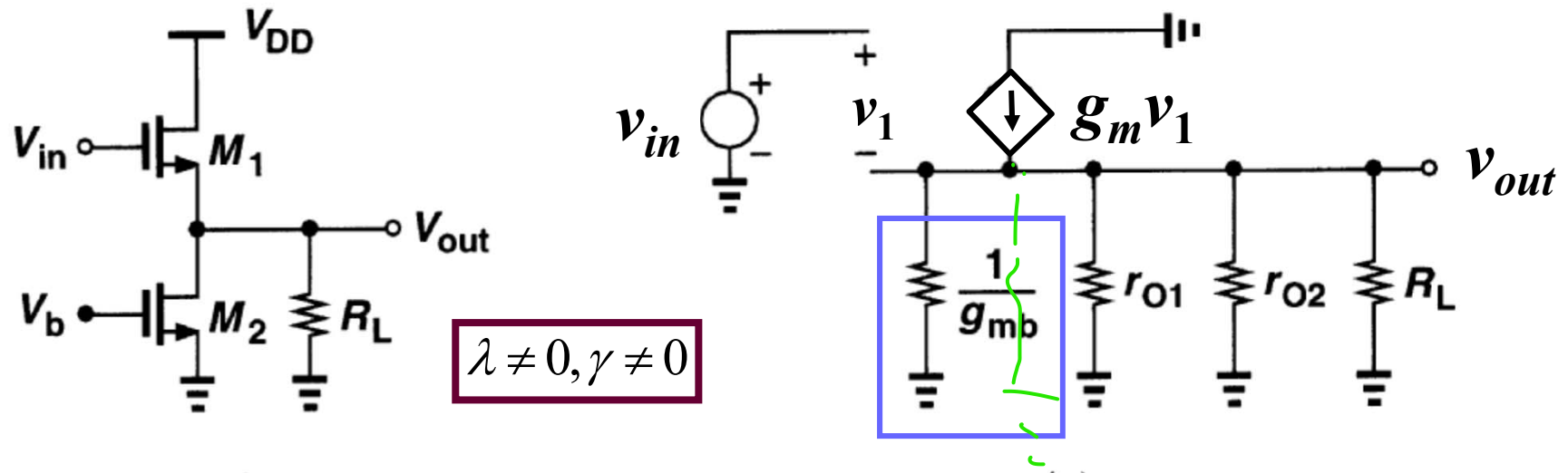
A low output resistance is obtained!

Modification of the small-signal circuit

- The body effect of M1 in source follower can be modeled as a resistor, simplifying analysis
 - Completed **linear v/i relationship**



Use source follower to drive finite R_L

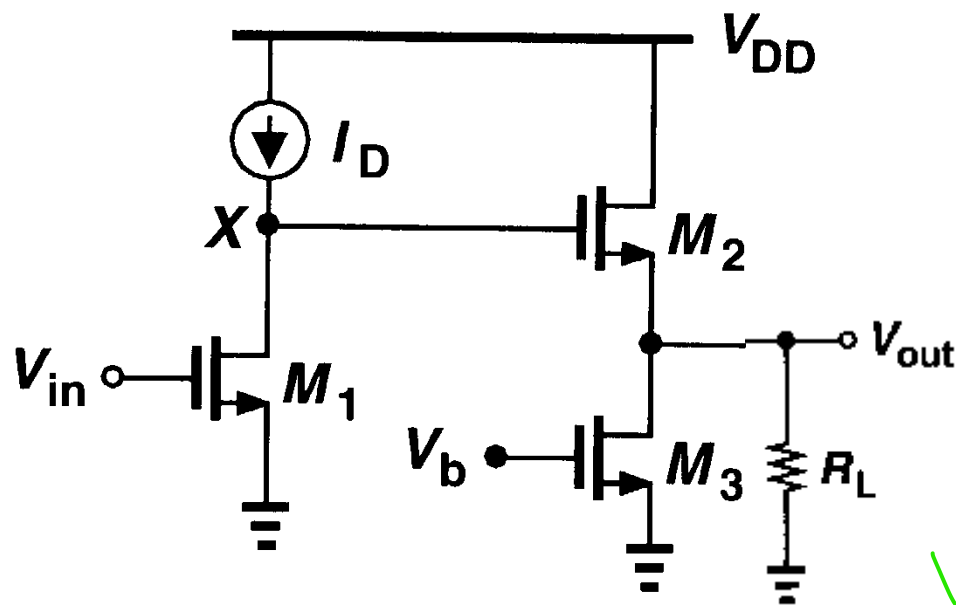


$$A_v = \frac{\frac{1}{g_{mb}} \parallel r_{O1} \parallel r_{O2} \parallel R_L}{\frac{1}{g_{mb}} \parallel r_{O1} \parallel r_{O2} \parallel R_L + \frac{1}{g_m}} \approx \boxed{\frac{R_L}{R_L + \frac{1}{g_m}}}$$



Application of source follower

- Source follower has infinite input resistance and medium or low output resistance
- It is often used as a voltage buffer to drive low R_L



Without SF

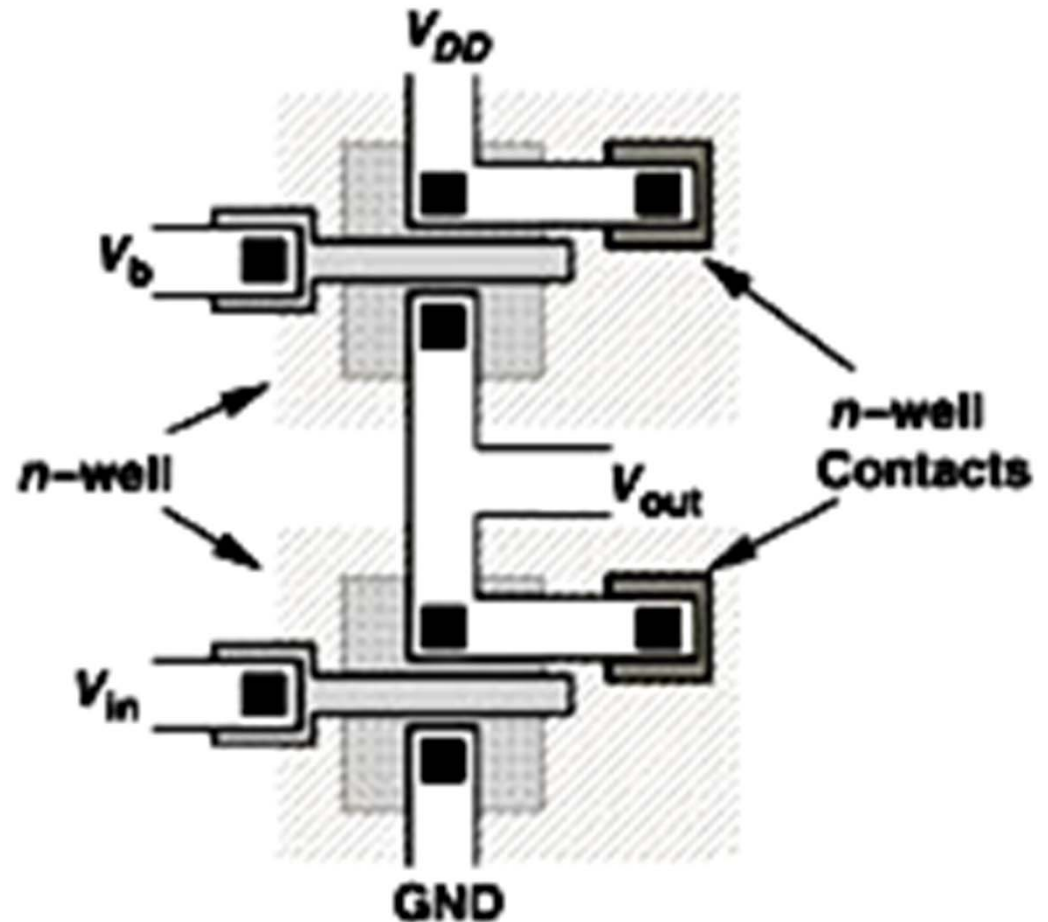
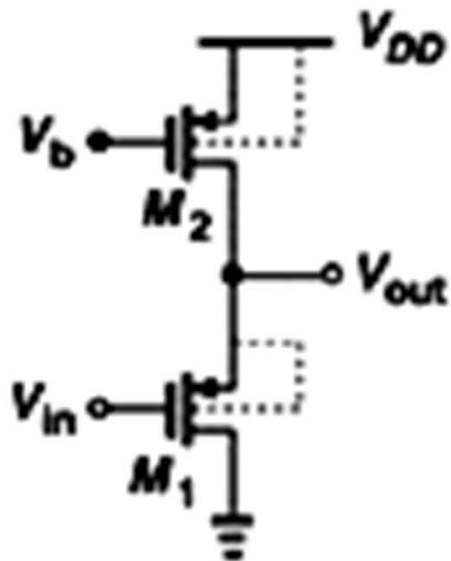
$$A_v = -g_{m1}R_L$$

With SF

$$A_v \approx -g_{m1}r_{o1} \cdot \frac{R_L}{R_L + 1/g_{m2}}$$



Use PMOS to Eliminate Nonlinearity



Outline

- Basic concepts
 - Common-Source stage
 - Source Follower
 - **Common-Gate stage**
 - Cascode stage
- 



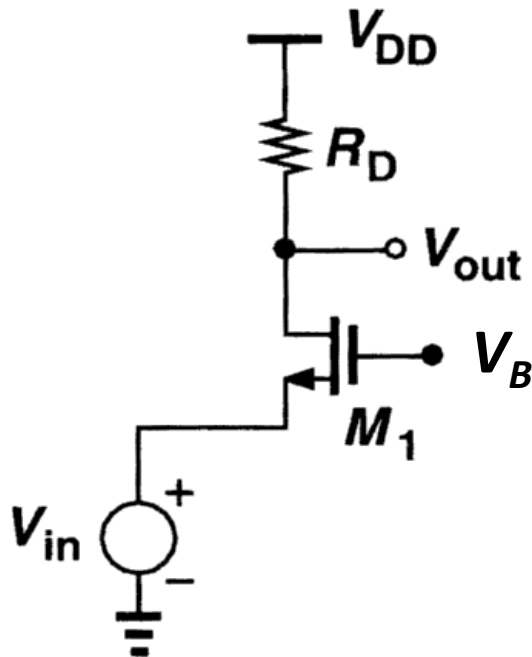
Common Gate

- The input resistance of CS stage is infinite
- However, in some applications, we need lower input resistance
 - Radio frequency (RF) systems need 50Ω input impedance
- Apply input signal into the source node can get lower input resistance
 - Common-gate (CG) stage



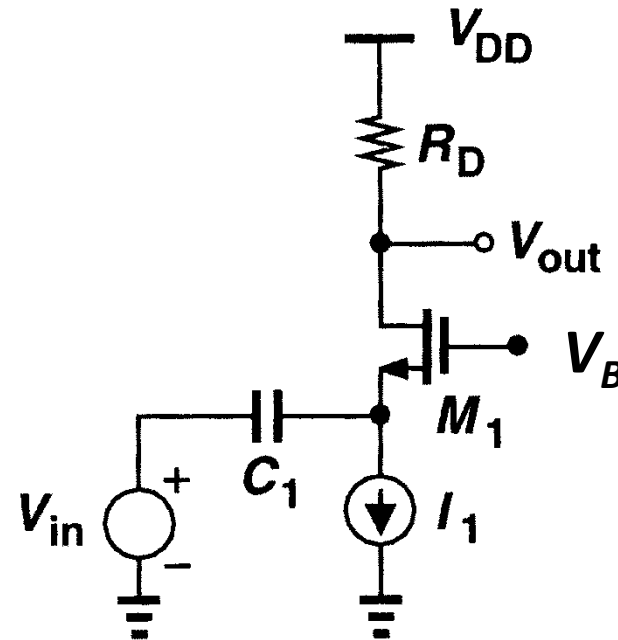
Common-Gate (CG) configuration

- direct coupling CG



Biasing current is determined both by V_B and V_{in}

- capacitive coupling CG



Biasing current is determined by I_1



Large signal analysis (direct coupling case)

- Consider that V_{in} decreases from a large value

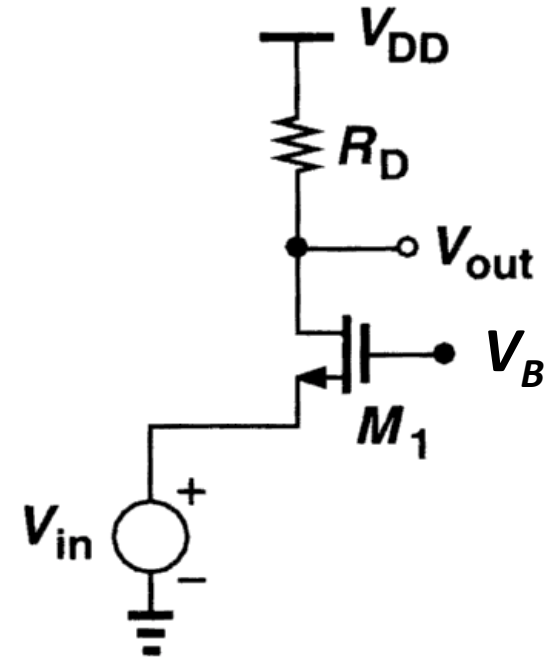
- When $V_{in} \geq V_B - V_{TH}$

- M_1 is off, $V_{out} = V_{DD}$

- For Lower V_{in}

$$I_d = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_B - V_{in} - V_{TH})^2$$

$$V_{out} = V_{DD} - \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_B - V_{in} - V_{TH})^2 \cdot R_D$$

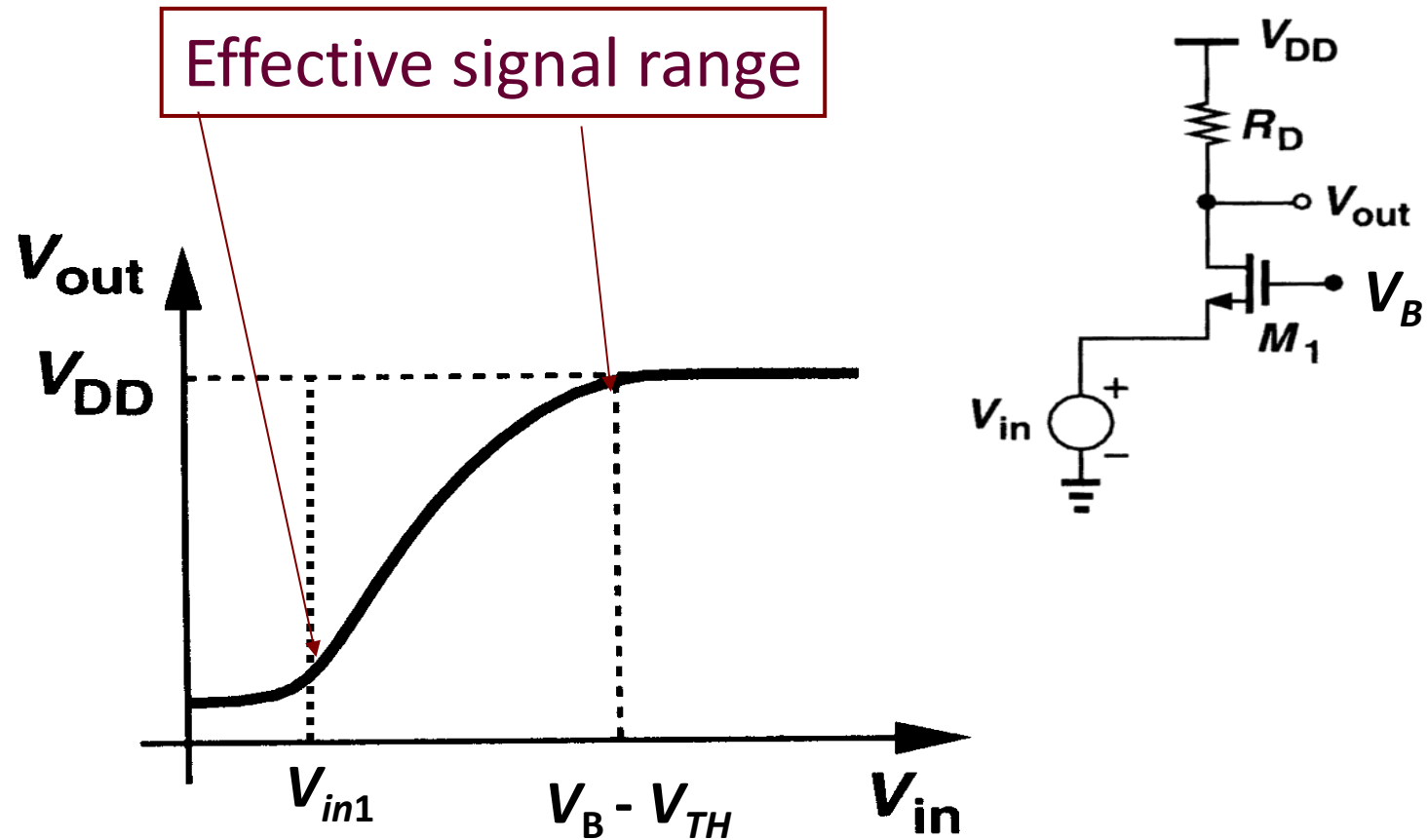


- When $V_{in} < V_{in1}$, M_1 enters triode region

$$V_{DD} - \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_B - V_{in1} - V_{TH})^2 \cdot R_D = V_B - V_{TH}$$



Large-signal characteristics



- Unlike CS stage, V_{out} increases with V_{in} !



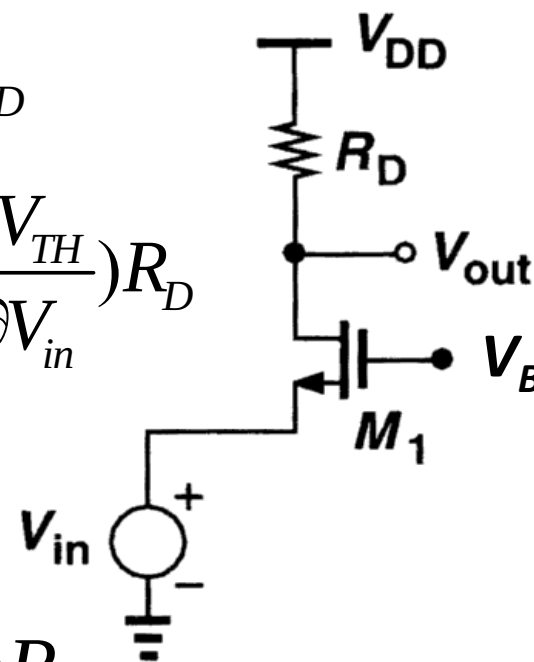
Finding A_v from large-signal analysis ($I=0$)

$$V_{out} = V_{DD} - \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_B - V_{in} - V_{TH})^2 R_D$$

$$\frac{\partial V_{out}}{\partial V_{in}} = -\mu_n C_{ox} \frac{W}{L} (V_B - V_{in} - V_{TH}) \left(-1 - \frac{\partial V_{TH}}{\partial V_{in}}\right) R_D$$

$$\therefore \frac{\partial V_{TH}}{\partial V_{in}} = \frac{\partial V_{TH}}{\partial V_{SB}} = \eta, \quad \therefore$$

$$A_v = \frac{\partial V_{out}}{\partial V_{in}} = \underbrace{\mu_n C_{ox} \frac{W}{L} (V_B - V_{in} - V_{TH})}_{g_m} (1 + \eta) R_D$$



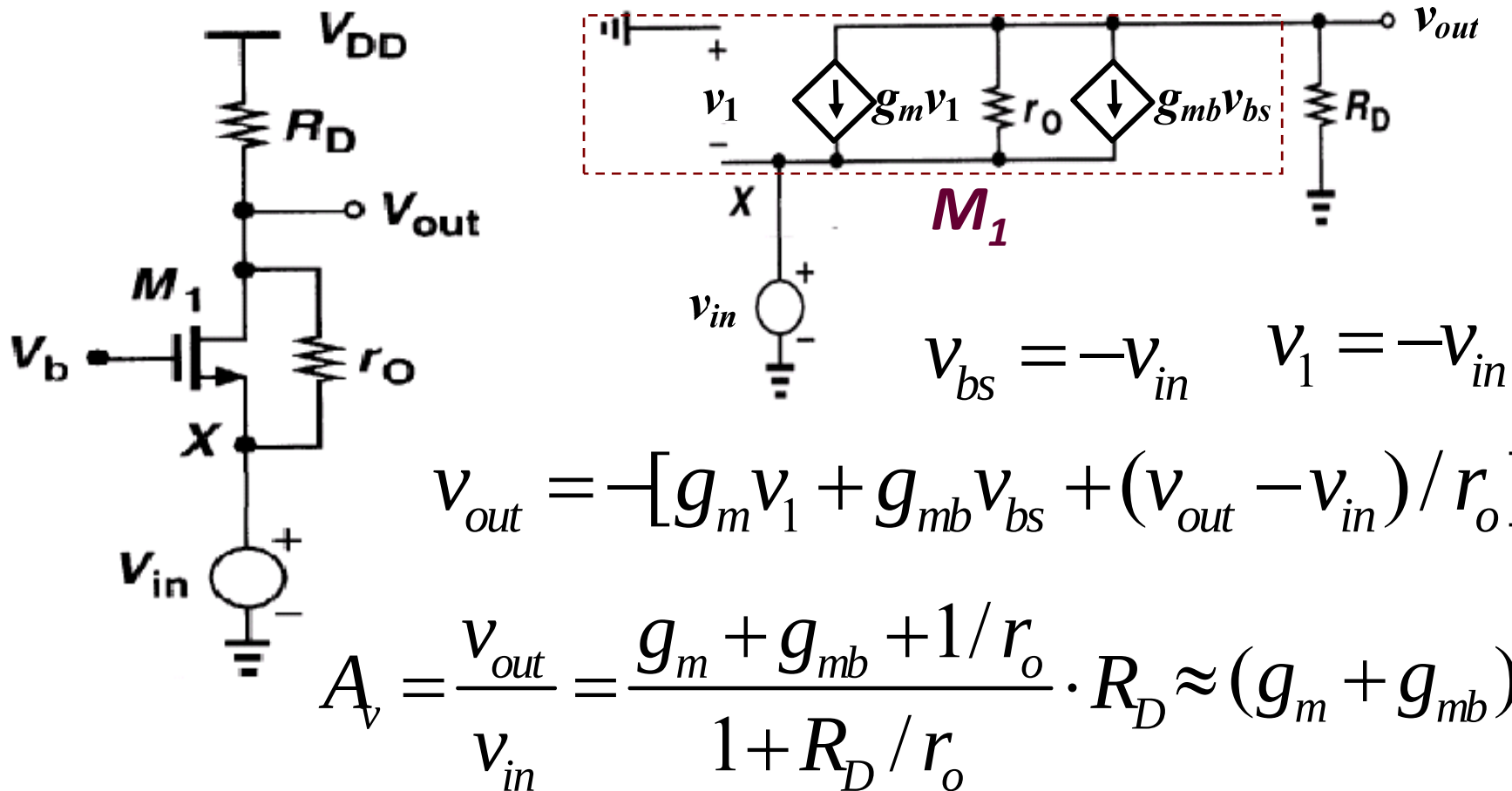
$$\rightarrow A_v = g_m (1 + \eta) R_D$$

- Gain is positive
- Body effect increases the equivalent transconductance!



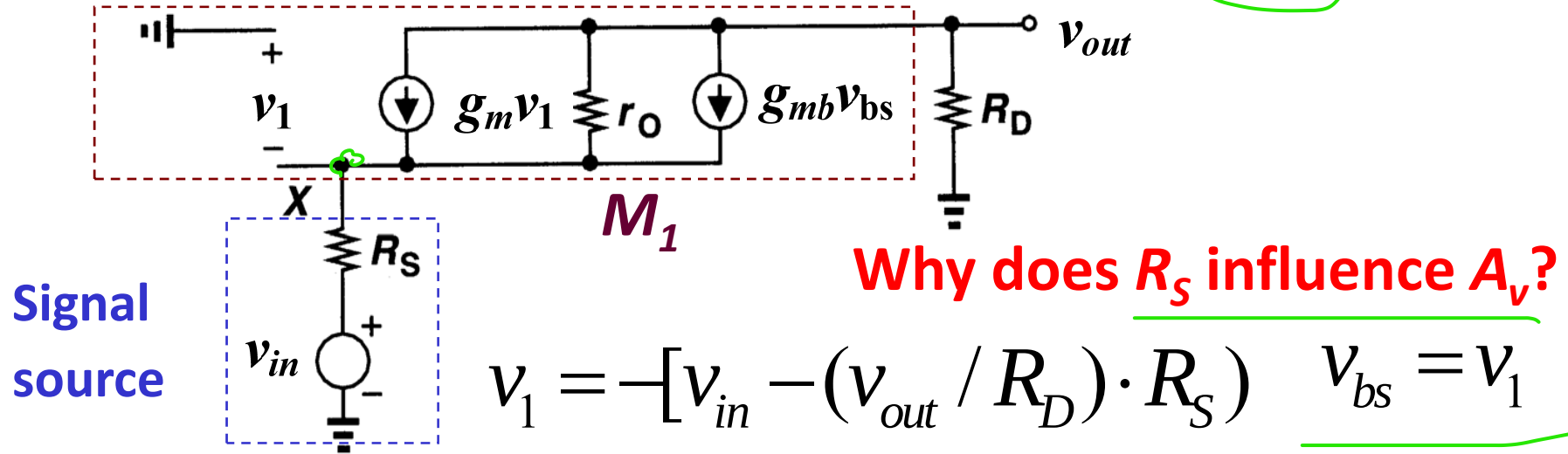
Finding A_v with Small-signal method

- Considering channel length modulation



Even more complex situation: source impedance

- Input signal source has finite impedance R_S

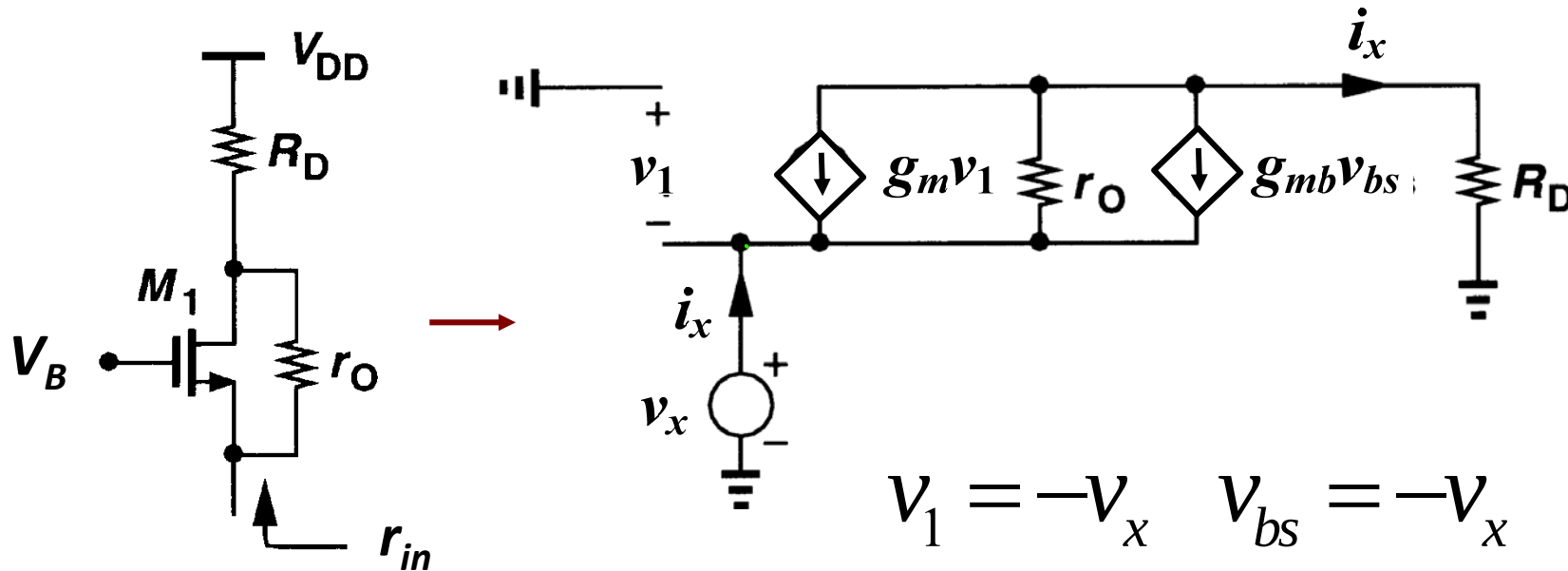


$$v_{out} = -[g_m v_1 + (v_{out} + v_1) / r_o + g_{mb} v_{bs}] \cdot R_D$$

$$A_v = \frac{v_{out}}{v_{in}} = \frac{g_m + g_{mb} + 1/r_o}{1 + (g_m + g_{mb})R_S + (R_D + R_S)/r_o} \cdot R_D$$



Input resistance of CG



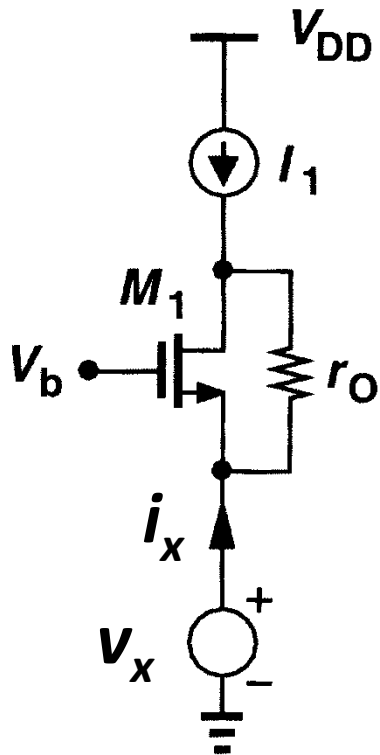
$$\underline{i_x R_D} + r_o (i_x - (g_m + g_{mb}) v_x) = v_x$$

$$\rightarrow r_{in} = \frac{v_x}{i_x} = \frac{R_D / r_o + 1}{1 / r_o + (g_m + g_{mb})} \approx \frac{1}{g_m + g_{mb}} \quad \text{When } R_D \text{ is small}$$



Input resistance is a function of R_D

- Considering the CG with a current source load



$$r_{in} = \frac{v_x}{i_x} = \frac{R_D / r_o + 1}{1 / r_o + (g_m + g_{mb})}$$

$$R_D = \infty$$

$$\rightarrow r_{in} = \infty !!$$



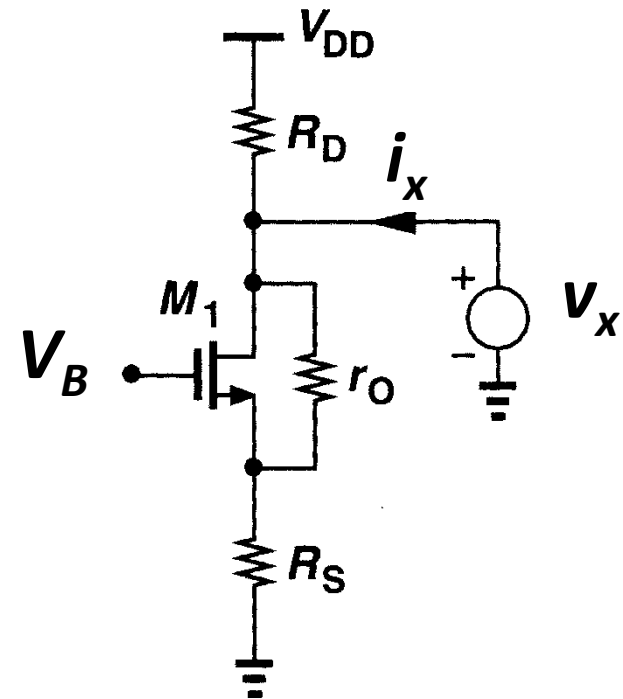
Output resistance

- Take the resistance of source into account
 - What does the circuit look like?
 - Yes, it is a CS stage with degeneration

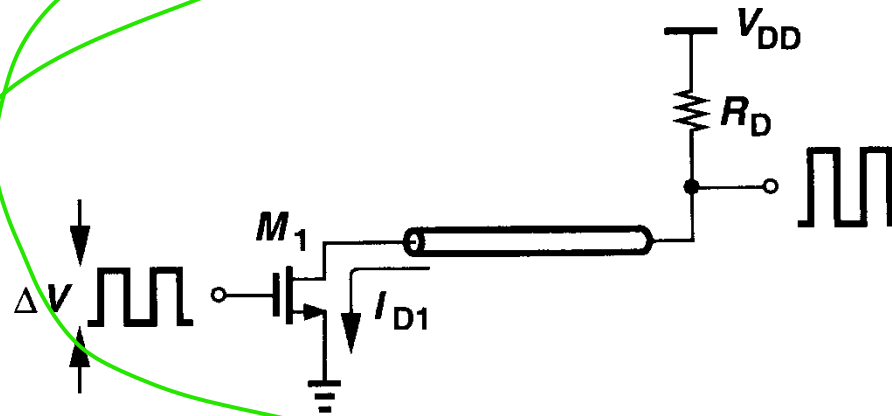
$$r_{out} = r_{OD} \parallel R_D$$
$$= \{r_O + [1 + (g_m + g_{mb})r_O] \cdot R_S\} \parallel R_D$$

– if $R_S=0$

→ $r_{out} = r_O \parallel R_D$ ✓



Application of CG stage (1)



- **Example 3.10**

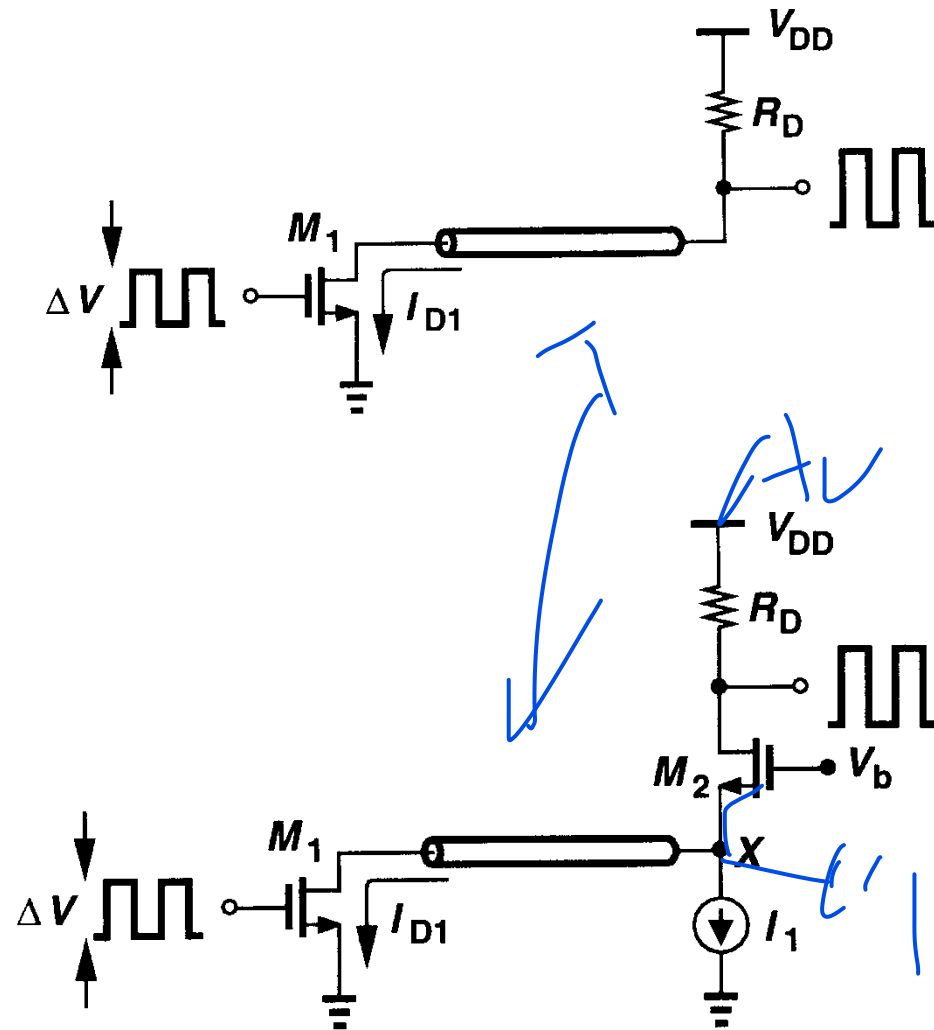
- Drive R_D through a **transmission line**
- The **intrinsic impedance** of most T-line is **50 Ω**
- To avoid reflection, termination impedance must equal to 50Ω (will study in high-freq. electronics)



Application of CG stage (2)

- Example 3.10

- A_v of both circuits are $g_m R_D$
- However, R_D in the CS stage must be 50Ω to avoid reflection, resulting in low gain
- The CG case can avoid reflection because of its low input resistance
- R_D in CG stage can be larger to obtain higher gain



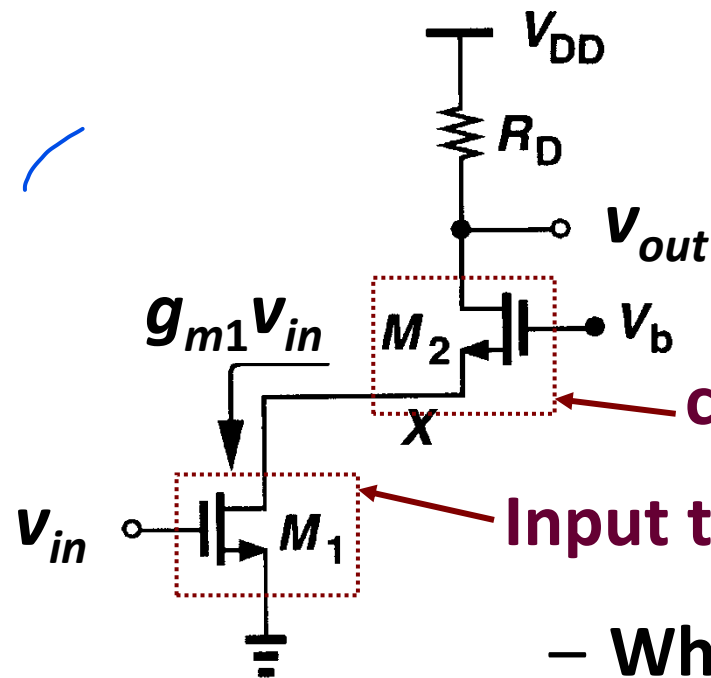
Outline

- Basic concepts
- Common-Source Stage
- Source Follower
- Common-Gate Stage
- **Cascode Stage**



Basic cascode configuration

- From EX. 3.10, we know that input of a CG stage can be current
- CS stage can turn voltage into current
- The *cascode* of CS and CG is called **cascode**



- An amplifier using 2 transistors
- M_1 and M_2 carries equal current

Input transistor

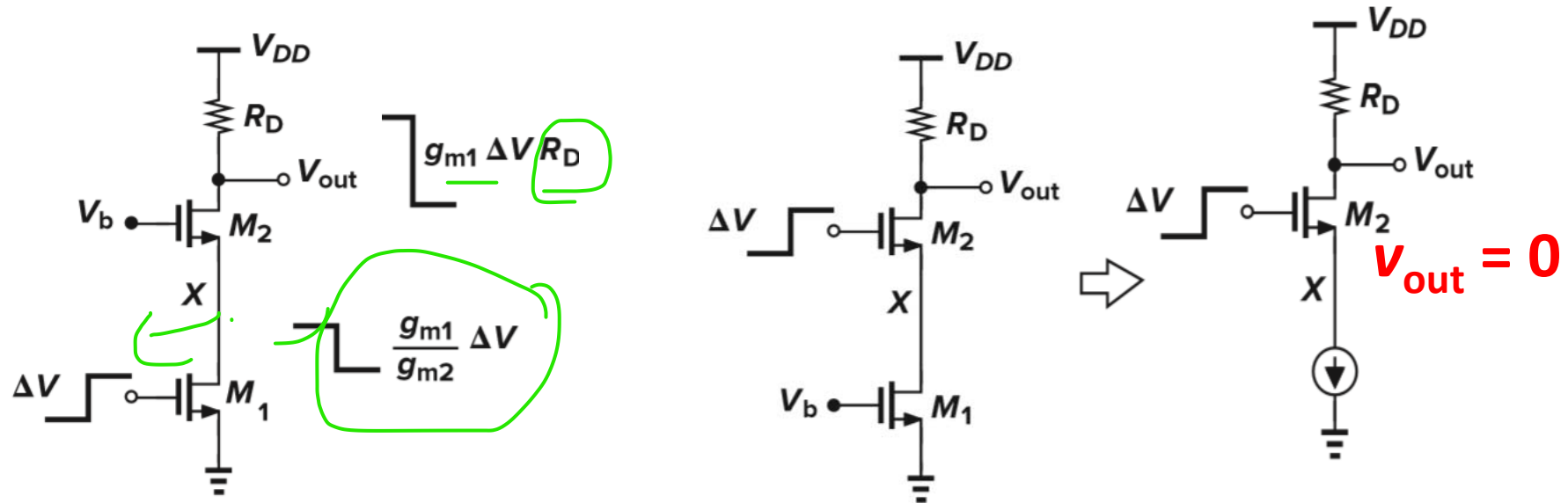
cascode transistor

– Why cascode?

Many advantages!!



Inspection: which is the input?



- Which one can be an amplifier?
 - M1 is the input transistor



Large signal range analysis

- M_1 and M_2 are both required to be saturated

- To ensure M1 saturation

$$V_x \geq V_{in} - V_{TH1} \quad V_x = V_B - V_{GS2}$$

$$\rightarrow V_B \geq V_{in} - V_{TH1} + V_{GS2}$$

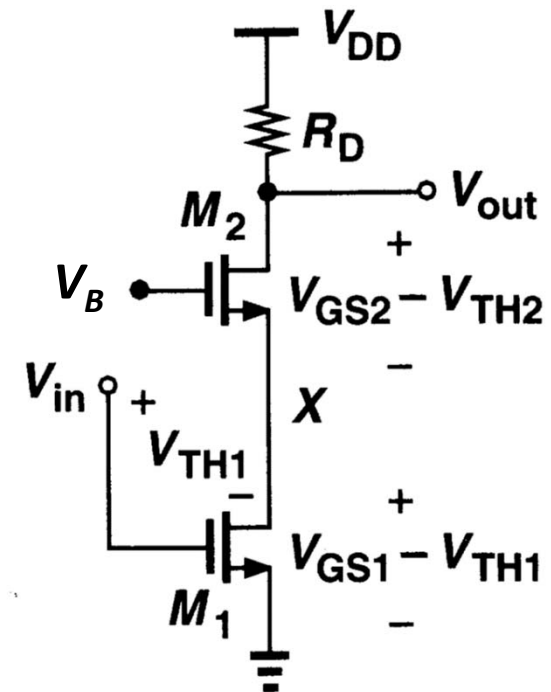
- For M2

$$V_{out} \geq V_B - V_{TH2}$$

$$\rightarrow V_{out} \geq V_{in} - V_{TH1} + V_{GS2} - V_{TH2}$$

$$V_{out} \geq V_{OD1} + V_{OD2}$$

– Output range is reduced by one V_{OD} !

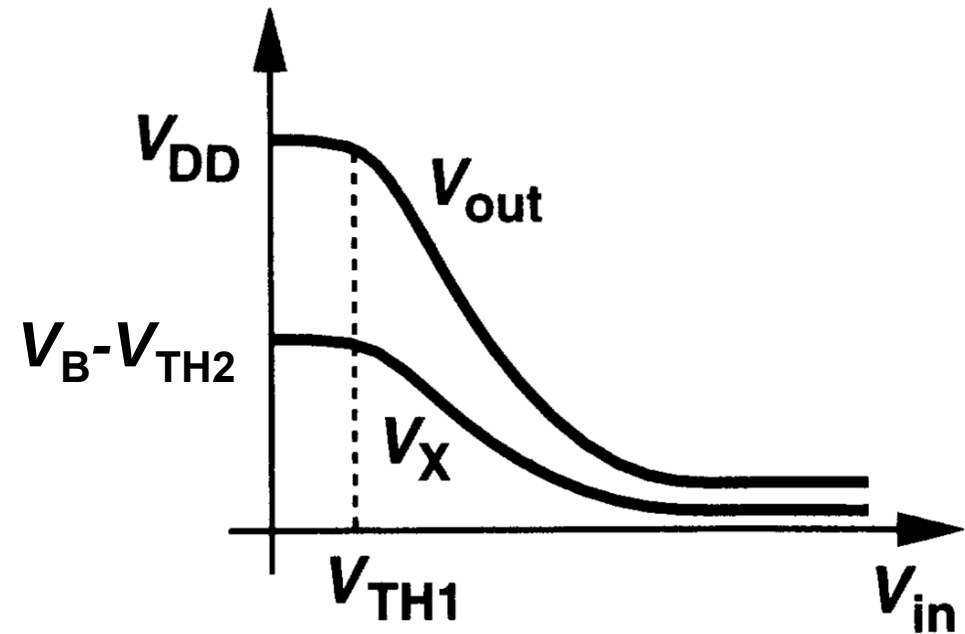


Input/output characteristics

- When $V_{in} < V_{TH1}$, M_1 and M_2 are off,

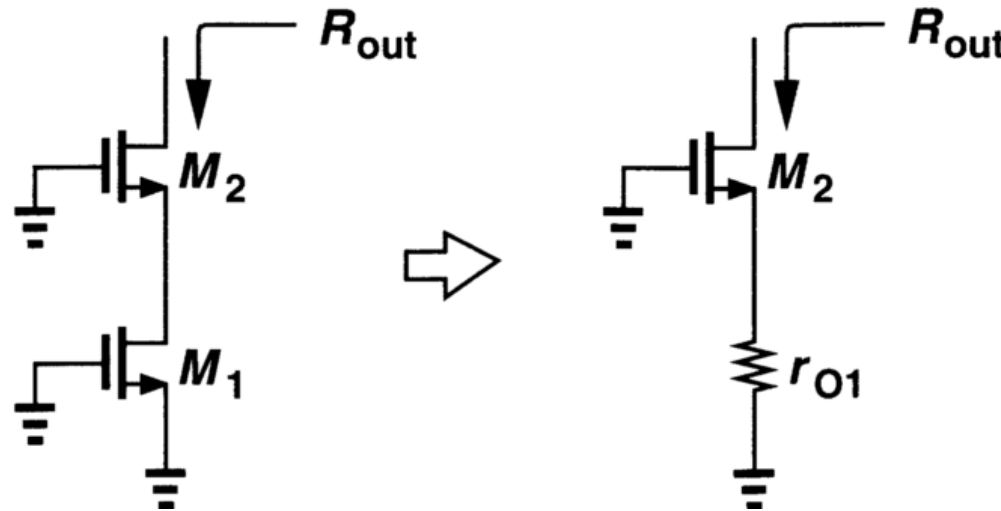
$$V_{out} = V_{DD}, V_X = V_B - V_{TH2}$$

- When $V_{in} > V_{TH1}$, V_{out} drops,
 - V_{GS2} increases, resulting in a drop of V_X
- If V_{in} increase further
 - Who enters triode region first?



Output Resistance of Cascode Stage

- Using the results of CS stage with degeneration:



$$R_{out} = [1 + (g_{m2} + g_{mb2})r_{O2}]r_{O1} + r_{O2}$$

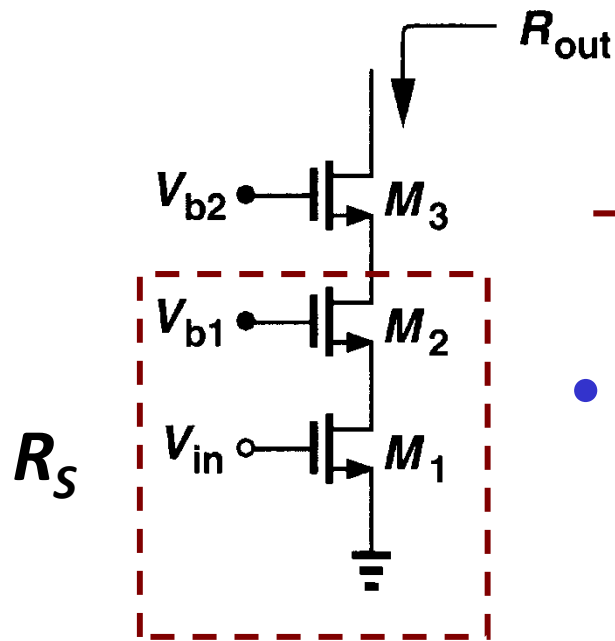
$$= r_{O1} + r_{O2} + (g_{m2} + g_{mb2})r_{O2}r_{O1} \quad \text{Important result!}$$

If $g_m r_O \gg 1$, then $R_{out} \approx (g_{m2} + g_{mb2})r_{O2}r_{O1} \approx g_{m2}r_{O2}r_{O1}$

Cascoding more transistors

- Still can use the results of CS with degeneration

$$R_{out} \approx (g_{m3} + g_{mb3})r_{O3}R_S \approx g_{m3}r_{O3}R_S$$



$$R_S \approx g_{m2}r_{O2}r_{O1}$$

$$\longrightarrow R_{out} \approx g_{m3}r_{O3} \cdot g_{m2}r_{O2} \cdot r_{O1}$$

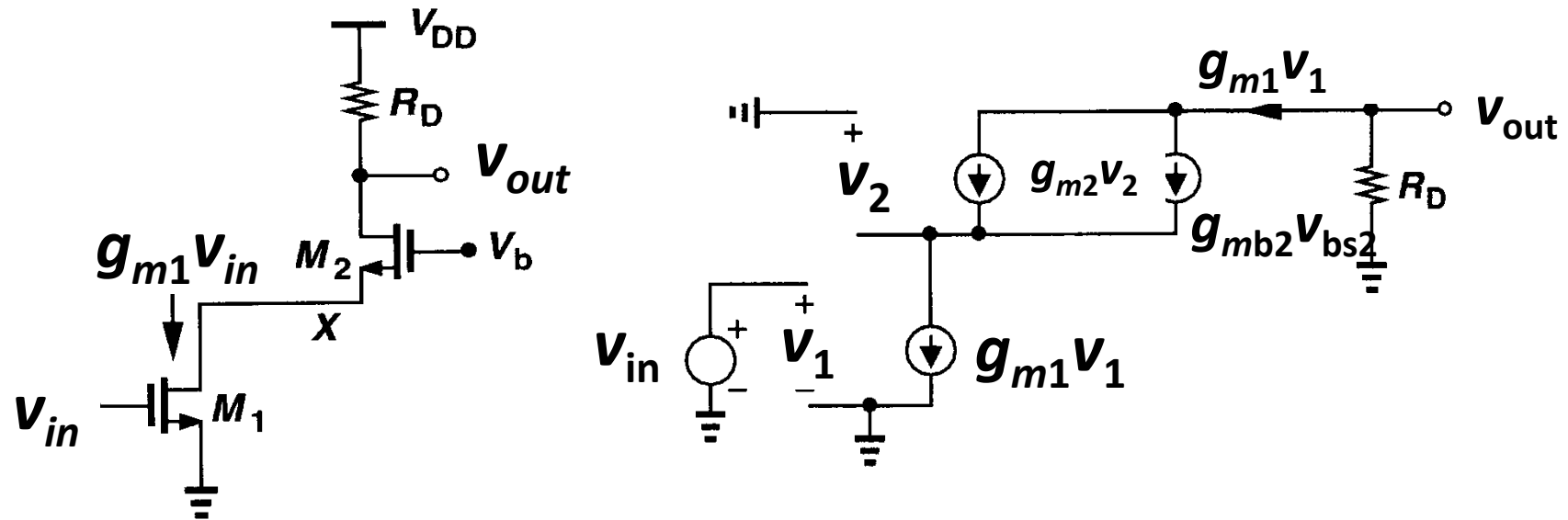
- A very large output resistance is obtained!

– The price is reduced signal range

$$V_{out,min} \geq V_{OD1} + V_{OD2} + V_{OD3}$$



Small-signal gain of cascode stage



- Neglecting the effect of λ
 - M_1 converts input voltage into currents
 - All currents flows into R_D

$$A_v = -g_{m1}R_D$$

- Same results with CS stage
 - What are the benefits?

Intrinsic gain of cascode stage

- Cascode with current source load

$$A_v = -G_m r_{out}$$

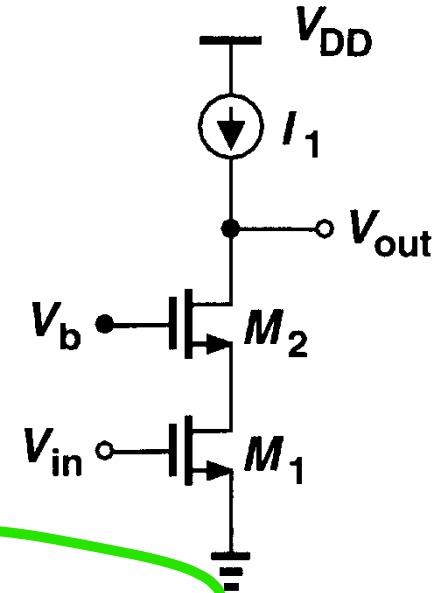
- Considering λ , G_m of circuit is a little bit smaller than g_{m1} , why?
(see Example 3.22)

- We still can assume $G_m \approx g_{m1}$

$$A_v \approx -g_{m1} \cdot g_{m2} r_{o2} r_{o1} = -g_{m1} r_{o1} \cdot g_{m2} r_{o2}$$

- The intrinsic gain is about the square of CS stage!**

- Exact derivation is given in Example 3. 22,
- pls. learn it by yourself.



Example 3.21

- Derive the voltage gain

– Key point: r_x

$$r_x = r_{o1} \parallel \left(\frac{1}{g_{m2} + g_{mb2}} \parallel r_{o2} \right)$$

$\mathbf{V_x}$

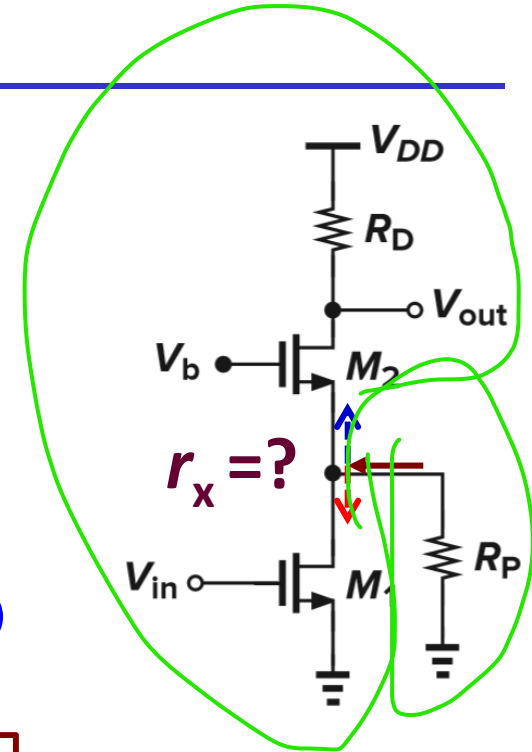
$$V_{out} = \boxed{v_{in} g_{m1} \cdot (r_x \parallel R_P)} \cdot (g_{m2} + g_{mb2}) \cdot (R_D \parallel r_{od})$$

$\mathbf{G_m}$

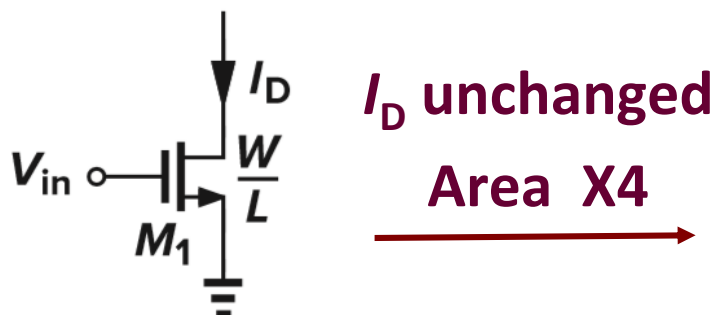
$$V_{out} = v_{in} g_{m1} \frac{r_{o1} \parallel r_{o2} \parallel 1/(g_{m2} + g_{mb2}) \parallel R_P}{\boxed{1/(g_{m2} + g_{mb2})}} \cdot (R_D \parallel r_{od})$$

$\mathbf{G_m}$

$$\lambda=0 \rightarrow \frac{V_{out}}{V_{in}} = - \boxed{g_{m1} \frac{(g_{m2} + g_{mb2}) R_P}{1 + (g_{m2} + g_{mb2}) R_P}} R_D$$



Two methods to improve the output impedance

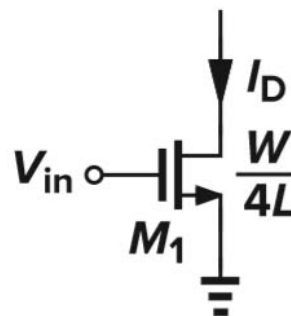


$$g_m = \sqrt{2I_D \mu_n C_{ox} \frac{W}{L}}$$

$$r_o = \frac{1}{\lambda I_D}$$

$$A_{int} = g_m r_o$$

- Method 1
increase L

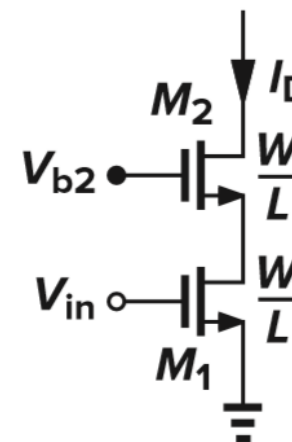


$$g_{m1} = \frac{g_m}{2}$$

$$r_{o1} = 4r_o$$

$$A_{int1} = 2A_{int}$$

- Method 2
cascode



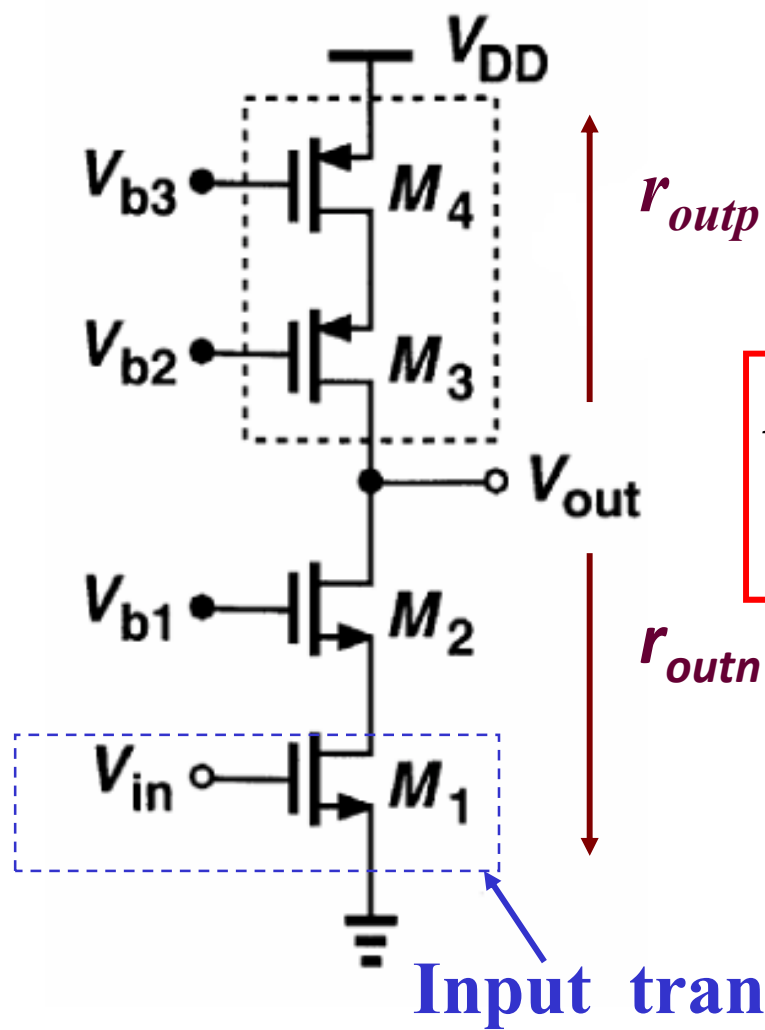
$$g_{m2} = g_m$$

$$r_{o2} = A_{int} r_o$$

$$A_{int2} = A_{int}^2$$



Use PMOS cascode current source as load



$$r_{outn} \approx g_{m2} r_{o2} \cdot r_{o1} \quad r_{outp} \approx ?$$

$$r_{outp} \approx g_{m3} r_{o3} \cdot r_{o4}$$

$$A_v \approx -g_{m1} \cdot (r_{outn} \parallel r_{outp})$$

$$\approx -g_{m1} \cdot (g_{m3} r_{o3} \cdot r_{o4} \parallel g_{m2} r_{o2} \cdot r_{o1})$$

$$V_{out\max} \leq V_{DD} - V_{OD3} - V_{OD4}$$

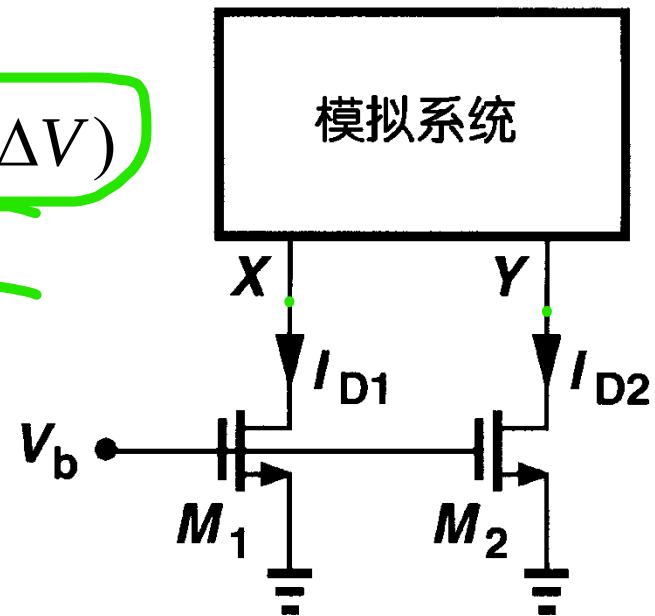
$$V_{out,min} \geq V_{OD1} + V_{OD2}$$



Understanding the shielding property of cascode

- Considering current source array case
 - M1 and M2 are designed with the same size
 - Ideally, $I_{D1} = I_{D2}$
 - $V_X - V_Y = \Delta V$ because of the analog system

$$I_{D1} - I_{D2} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_b - V_{TH})^2 (\lambda \cdot \Delta V)$$



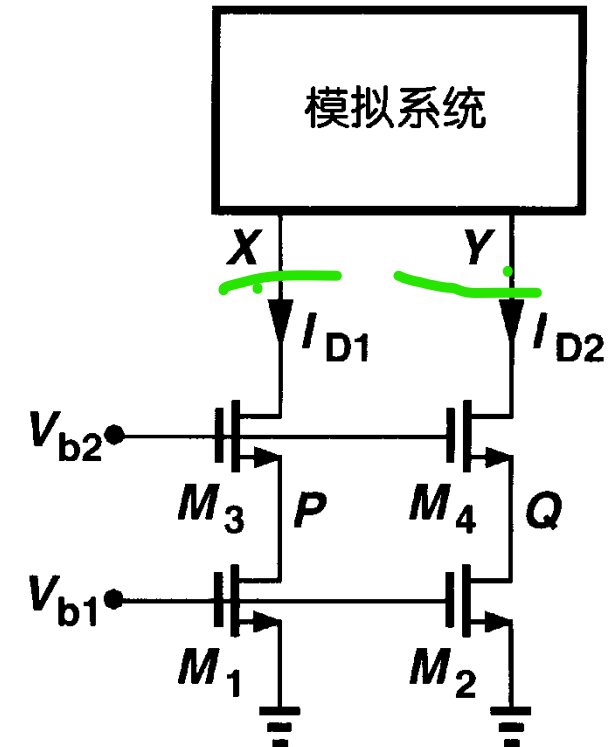
Cascode case

- $V_X - V_Y = \Delta V$

$$\Delta I = \frac{\Delta V}{(g_{m3,4} + g_{mb3,4})r_{o3,4} \cdot r_{o1,2}}$$

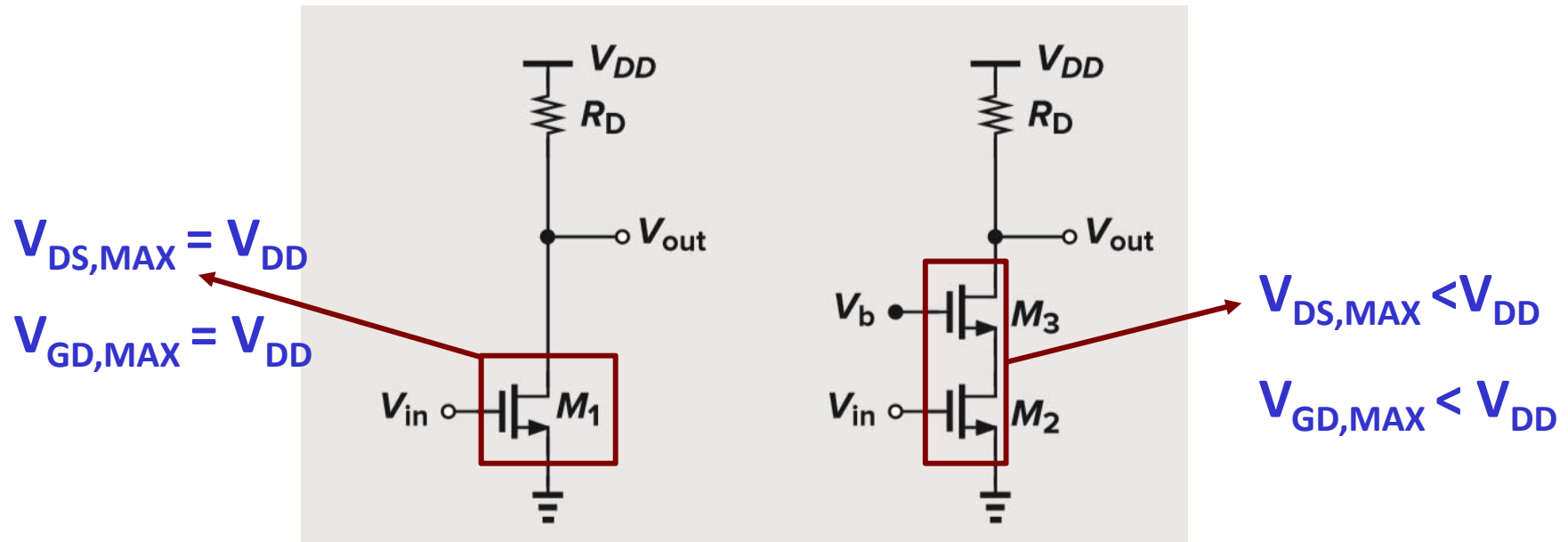
$$\Delta V_{P,Q} = \Delta I \cdot r_{o1,2} = \frac{\Delta V}{(g_{m3,4} + g_{mb3,4})r_{o3,4}}$$

$$I_{D1} - I_{D2} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_b - V_{TH})^2 \cdot \frac{\lambda \cdot \Delta V}{(g_{m3,4} + g_{mb3,4})r_{o3,4}}$$



- The variations at node X/Y are shielded by M3,4 !**

Cascode's additional advantage in nanometer design

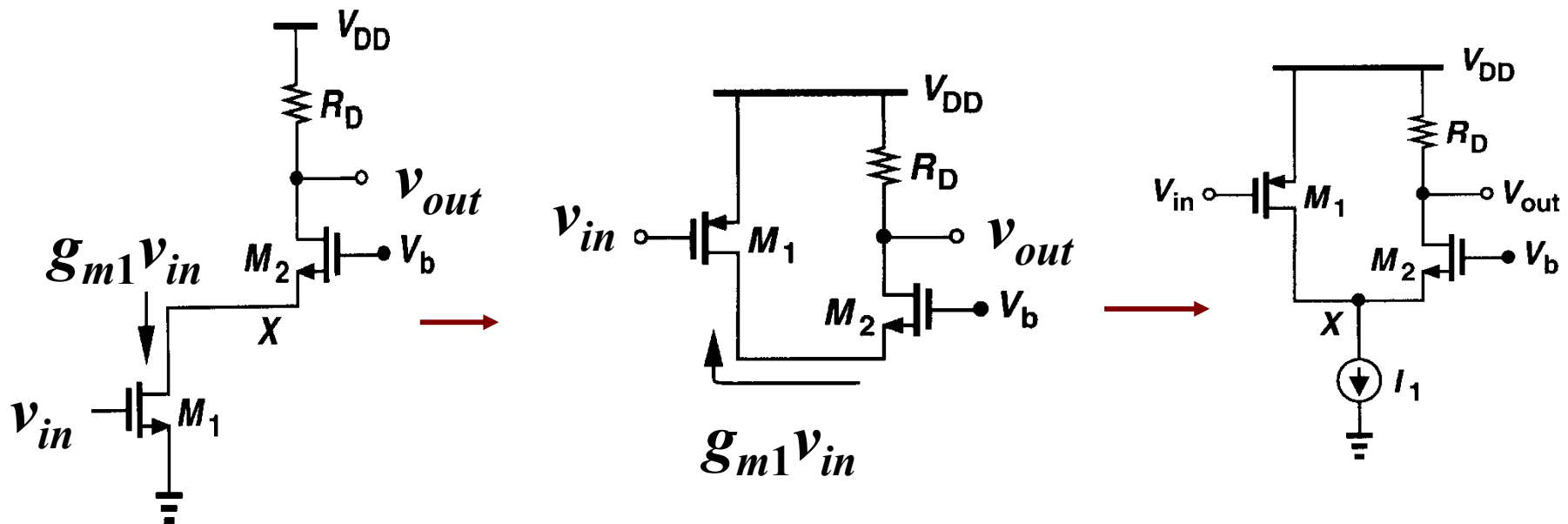


- In nanometer design, V_{GS} , V_{DS} , V_{GD} should be lower than a given value to avoid excessive stress, for a CS stage, V_{DD} must be kept lower than a given value
- In some design, we want to use a higher V_{DD} to increase the signal range, and hence the SNR
 - Cascode is a preferred option

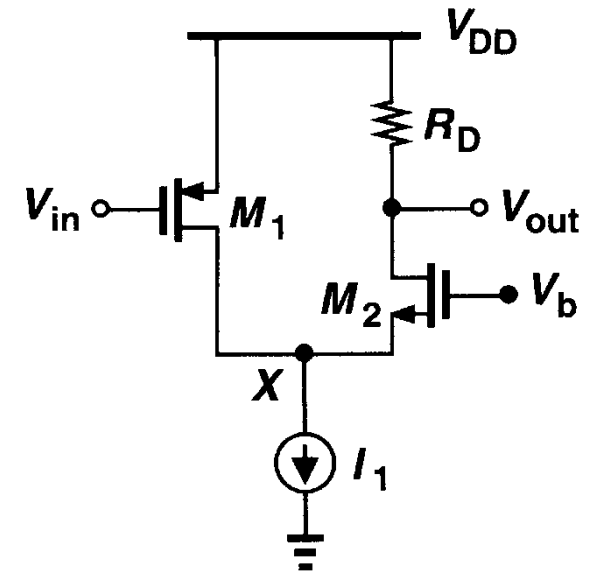
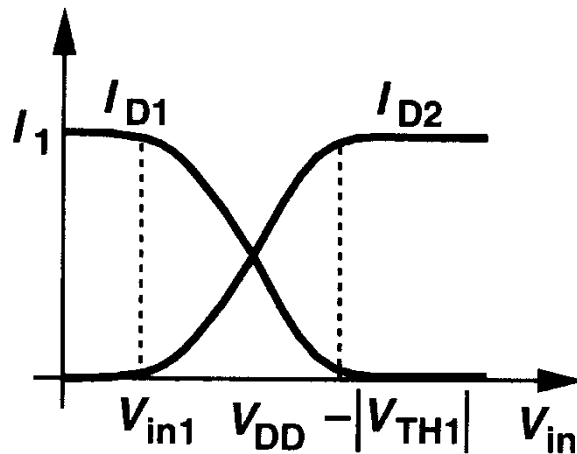
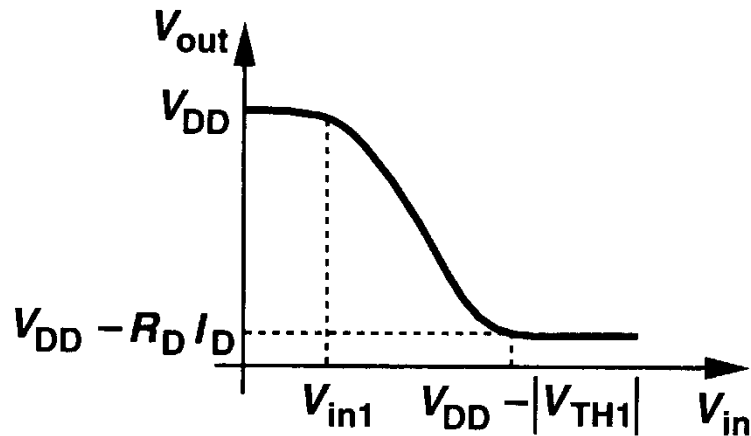


Folded cascode stage

- Problem of cascode
 - The input voltage is limited by the bias of the cascode transistor
- Use folded cascode to enlarge the input range



Large-signal behavior



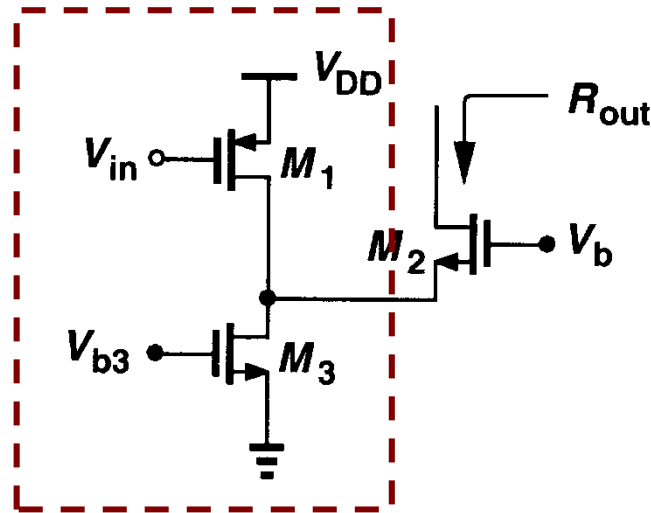
$$V_{in,max} \leq V_{DD} - |V_{TH1}|$$

$$V_{in,min} > V_{in1}$$



Output impedance of folded cascode

- Still use the results of CS with degeneration



$$R_S \approx r_{o1} \parallel r_{o3}$$

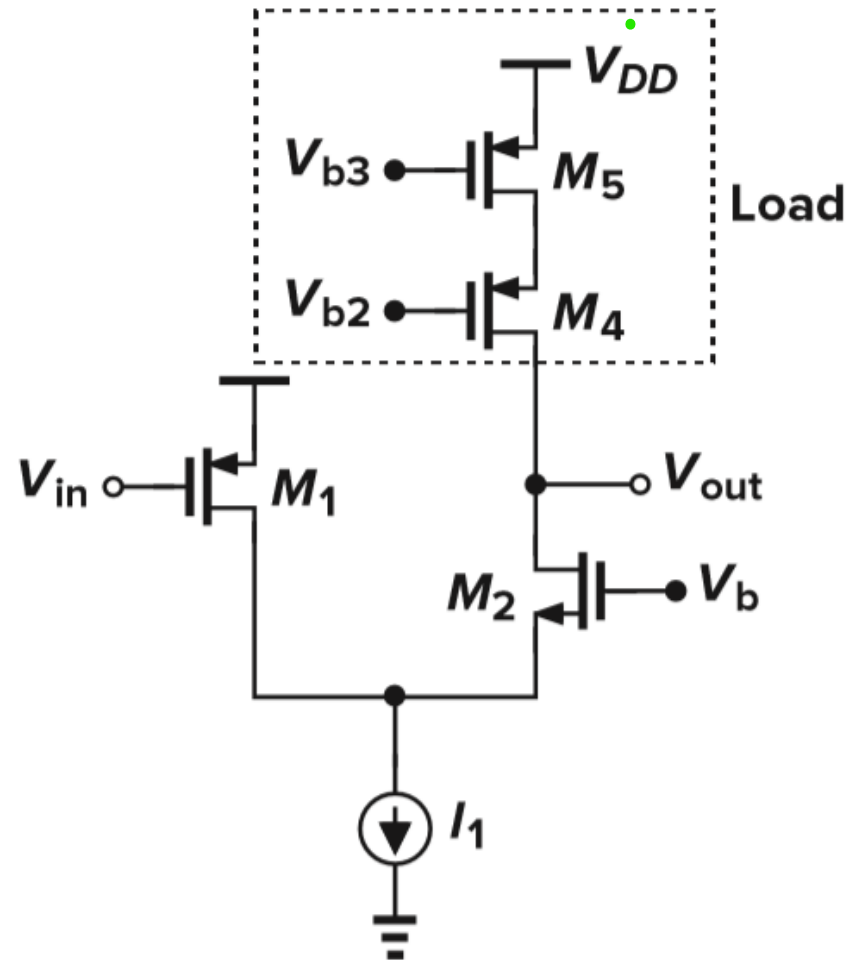
$$R_{out} = [1 + (g_{m2} + g_{mb2})r_{O2}] \cdot (r_{O1} \parallel r_{o3}) + r_{O2}$$

$$\approx g_{m2}r_{O2} \cdot (r_{O1} \parallel r_{o3})$$



Folded-cascode stage with a cascode load

- Use a PMOS cascode as the load to boost the gain
- Please write down the gain expression by yourself.



summary

- Basic concepts
- Common-Source stage
 - Resistive load
 - Diode-connected load
 - Current source load
 - Versatile analysis method for CS stage
 - CS stage with source degeneration
- Source Follower
- Common-Gate stage
- Cascode stage